

Frescox

I.J. Thompson
Nuclear Theory and Modeling Group
PO Box 808, L-414, Livermore, CA 94551, USA
Telephone + 1 (925) 423 4978

Email: I-Thompson@llnl.gov
Available: <https://github.com/LLNL/Frescox>
More information: <http://www.fresco.org.uk>

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Contents

1	Introduction	3
2	Input card-image format	4
3	Input NAMELIST format	4
3.1	Initial Lines	4
3.2	The &FRESCO namelist	4
3.2.1	Radial coordinates	4
3.2.2	Partial waves	5
3.2.3	Angular distributions	6
3.2.4	Defining coupled equations	6
3.2.5	Incident channel	8
3.2.6	Solving equations	8
3.2.7	R-matrix setup	9
3.2.8	Trace variables	10
3.2.9	Output details	10
3.3	&PARTITION and &STATES namelists	14
3.3.1	&PARTITION namelist	14
3.3.2	&STATES namelists	15
3.4	&POT potential namelists	16
3.4.1	First kind of &pot namelist	16
3.4.2	Second kind of &pot namelist: TYPE>0	16
3.4.3	Deformations	19
3.4.4	Pairwise couplings	20
3.5	&OVERLAP namelists for one- and two-particle form factors	23

3.5.1	One-particle KINDs	25
3.5.2	Two-particle KINDs	26
3.5.3	Namelist &TWONT	27
3.6	&COUPLING namelists	28
3.6.1	&CFP namelists	34
4	Input CDCC format	36
5	Input SFRESCOX format	39
5.1	Search file	39
5.2	Sfrescox commands	42
6	Test run: a CCBA calculation	45
7	File allocations	46
8	KIND=1 Spin Transfer Couplings	48

1 Introduction

FRESCOX is a Coupled-Channels program that can have finite-range transfer interactions among any number of mass partitions, and any number of nuclear excitations in each partition.

This is accomplished by calculating and storing the kernels of the non-local interactions, and then solving the coupled-channels set iteratively. At each iteration the stored kernels are used to integrate the wave functions to generate the source terms for the next iteration. Pade approximants can be used if the iterations diverge because of strong couplings.

The program can also generate local couplings for the rotational or single-particle excitations of either the projectile or the target in any partition, the local form factors for zero-range or local-energy-approximation can also be used. These various local interactions may also be included iteratively (to give multistep DWBA), or alternatively the first few coupled channels may be blocked together and solved by exact coupled-channels methods.

Special treatment is provided for any long ranged Coulomb multipoles, using James Christley's coupled-Coulomb functions CRCWFN, and interpolation in partial waves is also possible.

The nonlocal kernels for single-particle transfers are calculated first at a much smaller number of interpolation points, and then expanded when necessary to calculate the source terms by integrating

$$S(R_f) = \int_0^{R_{match}} K_{fi}(R_f, R_i) u(R_i) dR_i \quad (1)$$

where RMATCH and HCM, the step size, are given in the &Fresco namelist. Since the kernel function $K_{fi}(R_f, R_i)$ is usually rapidly varying with $D_{fi} = R_f - R_i$ (especially with heavy-ion reactions), and only slowly varying with R_f (if D_{fi} is constant), FRESCOX calculates and stores the function $K'_{fi}(R_f, D_{fi})$ at intervals of RINTP in R_f , and intervals of HNL in D_{fi} . The D_{fi} range considered is CENTRE-RNL/2 to CENTRE+RNL/2, i.e. range of RNL centred at CENTRE, and FRESCOX later suggests improved values for RNL & CENTRE. The HNL reflects to physical variation of K'_{fi} with D_{fi} , and can be a fraction of HCM (for heavy ion reactions) or a multiple of HCM (for light ion reactions especially with 'prior' interactions). If HNL is a submultiple of HCM, the program anticipates the interpolation of $u(R_i)$ in the equation above, and only stores an effective kernel function at intervals of HCM.

The new variable MTMIN in the FRT version controls the method used to calculate the form factors of the transfer kernels. If the L-transfer value (estimated by $\ell_f + \ell_i$) is larger than or equal to MTMIN, then the m-dependent expressions for the spherical harmonics are used explicitly. Otherwise, the angular dependencies of the initial and final transfer states are transformed into those of the channel variables R_i & R_f using Moshinsky's solid-harmonic transformation, as given for example in Austern et al. Phys. Rev. **133** pp B 3 - 16. By careful control of the numerical approximations, the second method can be extended to find heavy-ion transfer form factors, but the maximum transferred angular momentum L is still limited by numerical cancellation errors arising from the finite roundoff errors in the computer. With 8-byte word lengths, the unit roundoff is approximately 10^{-15} , and this limits the transferred L to 6 or less, so the default value of MTMIN is set to 6. The Moshinsky method is much faster when it is accurate, so the default value of MTMIN has been set as high as practical.

A general description of the formalisms used is given in I.J. Thompson, *Coupled Reaction Channels Calculations in Nuclear Physics*, Computer Physics Reports, **7** (1988) pp 167 - 212.

2 Input card-image format

Please see document frescox-input-manual.pdf.

3 Input NAMELIST format

The namelist input has two fixed-format lines to start with, and these followed by the namelists in Fortran style free-format. Variable names may be upper or lower case without distinction, but in this manual they are all capitalized. All namelists are terminated by a /. Some systems trailing comments begun by a !.

3.1 Initial Lines

Header line

One line of text to describe the nature of this run, for user's information only. (120 characters max)

NAMELIST

Fixed string to indicate input format using namelists

3.2 The &FRESCO namelist

Complete list of variables

Radial coordinates: hcm, rmatch, rintp, hnl, rnl, centre, hnn, rnn, rmin, rsp, cutl, cutr, cutc, rasy, accrcy, switch, sinjmax, ajswtch,

Partial waves: jtmin, jtmax, absend, jleast, jump, jbord, pset, jset, iso, llmax,

Angular distributions: kqmax, thmin, thmax, thinc, pp, koords, nearfa, dist0,

Defining coupled equations: inh, nnu, maxl, minl, mtmin, epc, erange, dk, plane, elpmax, rela, relref, ccbins, complexbins, sumform, ompform, pluto, unitmass, finec,

Incident channel: pel, exl, lab, lin, lex, elab, nlab,

Solving equations: ips, it0, iter, fatal, iblock, pade, nosol, dry, smallchan, smallcoup, ccreal, psiren, initwf, mpihelp, maxcoup, expand,

R-matrix setup: nrbases, nlagcc, nrbmin, pralpha, pcon, rmatr, btype, bndx, buttle, weak, krm, eigens, meigs, vsearch, echan, enodes, nparameters, eobs

Trace variables: chans, listcc, treneg, cdetr, smats,

Output details: xstabl, nlpl, waves, lampl, veff, kfus, nfus, wdisk, bpm, melfil, cdcc, tcfil, tcfilename, dist0, tmp, grace, boldplot

3.2.1 Radial coordinates

HCM, RMATCH, RINTP, HNL, RNL, CENTRE, HNN, RNN, RMIN, RSP, CUTL, CUTR, CUTC

Wave functions calculated at intervals of HCM up to $\text{abs}(\text{RMATCH})$.

If $\text{RMATCH} < 0$, then use values of RASYM, ACCRCY, SWITCH, AJSWITCH for coupled Coulomb wave functions.

Non-local kernels $K'_{fi}(R_f, D_{fi})$ calculated at R_f intervals of RINTP, and for a non-local (D_{fi}) range of RNL centred at CENTRE in steps of HNL.

RMATCH and RINTP are rounded to multiples of HCM, and HNL is rounded either to a multiple or a sub-multiple of HCM.

For two-nucleon transfers, the nucleon-nucleon distance is discretized from RMIN to RNN in a multiple of 6 Gaussian quadrature points to give step size close to HNN.

RSP is the upper limit of state radius when folding single-particle states (bound states or continuum bins) with KIND=3 or 4 couplings.

CUTL: set the number of radial points per ℓ of the lower radial cutoff when integrating the radial equations. Default = -1.6

When $\text{CUTL} > 0$, use $\ell = J$ (total angular momentum of CC set),

When $\text{CUTL} < 0$, use $\ell = L_{in}$ (orbital angular momentum of incoming partial wave). Using $\text{CUTL} < 0$ gives more accurate analyzing powers.

CUTR = lower radial cutoff (fm). Combining with CUTL is to use $R_{\min} = \max(\text{CUTL} * \ell * \text{HCM}, \text{CUTR})$.

If $\text{CUTR} < 0$, put cutoff at point-Coulomb turning point $-|\text{CUTR}|$.

CUTC = lower radial cutoff (in fm) for off-diagonal couplings.

RASYM, ACCRCY, SWITCH, SINJMAX, AJSWITCH

Use coupled Coulomb wave functions from CRCWFN out to asymptotic radius RASYM from inner radius $\text{abs}(\text{RMATCH})$ for those partitions in which PWF is TRUE.

If $\text{RASYM} < 0$, then determine the outer radius in order that classical Coulomb trajectories reach an angle $\text{abs}(\text{RASYM})$ degrees.

ACCRCY is an accuracy parameter controlling the piecewise step length. Default is 0.01: smaller values give greater accuracy.

SWITCH is the radius at which to switch from Airy functions to sines and cosines in piecewise method. Default is 1000 fm. If $\text{SINJMAX} > 0$, then change switchover condition to $J > \text{SINJMAX}$.

AJSWITCH Normally the Coupled Coulomb wfns are matched to zero and the Numerov integration is omitted when the Coulomb distance of closest approach is more than 4.5 fm outside $\text{abs}(\text{RMATCH})$ (or the $-\text{CUTR}$ distance if CUTR negative). This is only allowed when $J \geq \text{AJSWITCH}$. Default is 0.0

3.2.2 Partial waves

JTMIN, JTMAX, ABSEND, JLEAST Calculate coupled-channels sets with total J in the interval $\max(0, \text{JTMIN}) < J < \text{JTMAX}$, stopping sooner if the absorption from the elastic channel is less than ABSEND millibarns of three successive J /parity sets. (If $\text{ABSEND} < 0$, this takes the full J interval.)

If $\text{JLEAST} > 0$, then only stop if $J > \text{JLEAST}$.

If $\text{JTMIN} < 0$, then for $J < \text{abs}(\text{JTMIN})$ include only the incoming channel in the calculations, ignoring transfers and excited states. This is needed if the elastic scattering cross sections are to be given correctly.

JUMP(1:), JBORD(1:) Calculate coupled-channels sets not for every J value, but at intervals of JUMP(i) for $J \geq \text{JBORD}(i)$, for $i=1,5$. (The program sets JUMP(0)=1 & JBORD(0)=JTMIN, and, if there are NJ non-zero values, also JBORD(NJ+1)=JTMAX. This gives no J jumping if JUMP and JBORD are not set in the namelist.) The omitted J values are provided by interpolation on the scattering amplitudes $A(m'M' : mM; L)$ prior to calculating cross sections.

PSET, JSET

If PSET = -1 or +1, restrict parity of CRC sets to that value (0 = no restriction)

If PSET = -2 or +2, restrict CRC sets to unnatural or natural parities respectively. Here, 'natural parity' is defined as parity = $(-1)^{nint(J+.1)}$.

JSET = number of CRC sets to calculate before stopping (0 = all sets)

ISO = 'A' or 'J', for replacing all barriers $L(L+1)$ by that for $L = J$ barrier

= 'B' or 'L', for replacing all barriers by $L = L_{in}$ barrier.

= 0 or blank, for no isocentrifugal approximations,

Of course, the *simplest and fastest* way to use the isocentrifugal approximation is to put all spins and parities to 0^+ , and all transition multipoles to $k = 0$. The ISO variable is not needed then.

LLMAX Maximum partial wave L in any CRC set.

3.2.3 Angular distributions

KQMAX, THMIN, THMAX, THINC Give cross sections (and tensor analyzing powers up to rank $K = \text{KQMAX}$) for centre-of-mass scattering angle from THMIN to $\text{abs}(\text{THMAX})$ in steps of THINC.

Elastic channels normally output the ratio to Rutherford, unless THMAX < 0.

PP Calculate analyzing powers/polarizations for projectile (PP=0 or blank), target (PP=1), ejectile (PP=2) or residual nucleus (PP=3). PP=4 gives projectile (PP=0) analyzing powers, along with Kyy results.

KOORDS determines the coordinate systems used for the analyzing powers:

= 0 : Madison coordinates (default)

= 1 : Madison + Transverse

= 2 : Madison + Transverse + Recoil

= 3 : Madison + Transverse + Recoil + Hooton-Johnson

NEARFA For mod(NEARFA,10) values:

= 0 or 1 for the usual cross sections,

= 2 or -2 for printing 'far side' cross sections too,

= 3 or -3 for printing far and near side cross sections too.

> 0 for printing far & near-sides for elastic channel only.

< 0 for printing far & near-sides for all channels.

If $\text{abs}(\text{NEARFA}) > 10$, also split the Coulomb amplitude according to Cha, CPC 176 (2007) 318.

3.2.4 Defining coupled equations

INH = 0 : zero-range transfer forms in intervals of HCM exactly

= 1 : stored in steps of HCM * (proj. core)/(proj. composite mass)

= 2 : stored in steps of HCM * (targ. core)/(targ. composite mass)

So INH=2 corrects for longitudinal recoil during transfers with zero-range projectiles.

NNU, MAXL, MINL, MTMIN, EPC control the accuracy of non-local transfer form factors:

NNU is the number of Gaussian integration points in the angular integration used for the non-local transfer kernels. NNU should be a multiple of 6; NNU = 18 is the minimum, and 24 or 36 give acceptable accuracy for all the reactions tried so far.

MAXL, MINL are the maximum and minimum L values for the non-local kernels. If zero, MAXL has the default value JTMAX+6, and if MINL < 0 it takes the default value |JTMIN|-6.

MTMIN is the lowest L -transfer for calculating transfer form factors using the m -dependent expressions for spherical harmonics. Putting MTMIN = 0 gives default value MTMIN = 6 (use MTMIN < 0 to avoid invoking default, if all transfers are to use this method).

EPC = percentage cutoff accuracy in the NNU angular integration. If zero, the default is $(30/\text{NNU})^2\%$.

ERANGE, DK set default parameters for continuum bins:

ERANGE = range of energies of the upper and lower boundaries of continuum bins:

if ERANGE > 0, then *ratio* of these energies;

if ERANGE < 0, then *difference* of the energies in MeV.

DK = step size of k (fm^{-1}) for integration over the ERANGE to construct the continuum bin.

PLANE, ELPMAX

PLANE = 1,3: zero Coulomb potential for elastic channel

= 2,3: zero Coulomb potentials for all nonelastic channels.

ELPMAX = energy ELAB below which the LPMAX limits in each partition are applied

RELA, RELREF $\neq$: use relativistic kinematics for the incident projectile:

If RELA contains 'a', use Ingemarsson eq(16) for kinematics

If RELA contains 'b', use Ingemarsson eq(17) for kinematics

If RELA contains 'h', use kinematics of G. Hale (LANL) with RELREF as the reference partition

If RELA contains 'f', use Ingemarsson eq(19) to scale potentials

If RELA contains 'g', use Ingemarsson eq(21) to scale potentials

If RELA contains 'r', get reduced mass from ratios of total energies

COMPLEXBINS, CCBINS, SUMFORM, OMPFORM

If COMPLEXBINS=T, allow for complex-valued bins (e.g. from optical potentials)

If CCBINS=T, allow for coupled-channels bins, which are necessarily complex-valued.

If SUMFORM = 1, 2 or 3: different methods for sum reductions of core-excitation bins.

OMPFORM = 0, 1, 2: different methods of parallel calculation of core-excitation bins.

PLUTO(:) = list of potential numbers KP to be prepared for the Lagrange-mesh method for single-particle eigenstates.

UNITMASS : unit (in amu) for MASS values read in. Default = 1.000

FINEC : $1/(\text{fine-structure constant})$: used to determine electrostatic e^2 . Default = 137.03599.

3.2.5 Incident channel

PEL, EXL, LAB, LIN, LEX Incoming plane waves are present in partition PEL with excitation pair EXL. The energies ELAB(:) are the laboratory energies for partition LAB's nucleus LIN (1 or 2 for projectile or target) in excitation pair LEX.

The defaults for PEL, EXL, LIN & LEX are all 1, and the default for LAB is PEL, so these variables can be normally omitted.

ELAB(:5), NLAB(:4) : Solve at different laboratory energies E until ELAB(i) = 0 is found.

If NLAB(i) > 1, then the range of E from ELAB(i) to ELAB(i+1) is covered in NLAB(i) linear intervals. For just one energy E , just specify 'ELAB = E '.

3.2.6 Solving equations

IPS, IT0, ITER, FATAL, IBLOCK, PADE Solve the coupled channels equations by at least IT0 iterations, and up to ITER iterations. Stop sooner if the absolute differences between successive S -matrix elements (scaled by $(2J+1)/(2.JTMAX+1)$) are less than IPS percent. (Excited state pairs with IGNORE set in the &STATE namelist are not counted against IPS).

Putting IT0=ITER zero solves only the elastic channel (along with the IBLOCK channels: see below). Putting IT0=ITER = 1 or 2 etc. gives 1 or 2-step DWBA.

Normally, a run is terminated if more than ITER steps are required for convergence.

Setting FATAL=False allows continuation even after convergence has failed after abs(ITER) iterations.

Iterations are normally also stopped if the successive differences are smaller than the errors estimated for the numerical integration of the coupled equations. Setting IPS < 0 uses abs(IPS), without this extra check.

IBLOCK is the number of pairs of excitation levels (starting from partition 1, excitation 1) that are coupled exactly by blocking together.

PADE = 0 for no Pade acceleration,

= 1 for Pade acceleration by the epsilon algorithm,

= 2 for Pade acceleration by finding the N/D polynomials.

NOSOL : if not to solve the CRC equations, only construct couplings

DRY normally F (false), but if T (true) the code does a 'dry run' to check that all arrays are of sufficient size. All coupled channel sets are generated, but only the elastic channels should be non-zero.

SMALLCHAN, SMALLCOUP, CCREAL

SMALLCHAN = fraction of unitarity to define a 'small channel'. A channel that is 'small' for NSMALL=5 times is dropped permanently.

SMALLCOUP: if all nonelastic channels are weaker than the fraction SMALLCOUP of unitarity, then permanently change from coupled-channels to DWBA.

CCREAL = T: assume all potentials and couplings real-valued

PSIREN : do simple renormalisation of channel wfs after Pade acceleration

INITWF : read in external scattering wave functions before iterating couplings. > 0: read in formatted data from file number INITWF.

< 0: read in unformatted data from file number INITWF.

The file format is that which is produced when WAVES > 0. All channels not read in are set to zero.

MPIHELP : number of OPENMP threads allocated to each MPI node

MAXCOUP(:), EXPAND(:) : fudge limits and factors (respectively) to adjust sizes of generated work arrays.

3.2.7 R-matrix setup

NRBASES

When NRBASES ≠ 0, R-matrix solutions are selected then all channels are ‘blocked’ together and solved in a full CRC procedure. All non-local potentials are included to all all orders (not iteratively).

NRBASES = target number of radial basis states in each channel. (Use 2*NRBASES for the elastic and first-inelastic channel for more accuracy).

If NRBASES < 0, then use *Lagrange mesh* basis with NLAG=–NRBASES basis functions.

NLAGCC > 0: another way to set NLAG = NLAGCC for a Lagrange mesh calculation.

NRBMIN, BUTTLE, PRALPHA, PCON, RMATR, BNDX(2), BTYPE, WEAK

NRBMIN = minimum number of radial basis states (default NRBASES).

BUTTLE = 4 for none, 0,2 for complex, 1,3 for real (2,3 without energy shift) Buttle correction. (default 0)

PRALPHA = print basis-state eigenvalues to files fort.60,61,62,63

PCON = trace variable for calculation of radial basis states (same meaning as IPC for single-particle bound states).

RMATR = R-matrix matching radius (default RMATCH). Warning: RMATR will be changed to an even multiple of HCM.

BNDX(i) is used to set the R-matrix Boundary Condition number depending on BTYPE. Use i=1,2 for positive, negative parity coupled channels sets (respectively).

Energy $\hbar^2 k^2 / 2m$, where $k = f' / f$, the logarithmic derivative for all radial basis states at $r=RMATR$, with k having the same sign as EBETA(i).

BTYPE = E,k, L,B, S : methods to calculate Bloch parameter B from BNDX. If BNDX(i) is the energy for 'E' or wave number for 'k', then B is inverse logarithmic derivative of Coulomb function there. Otherwise B = –L for 'L', B = BNDX(i) for 'B', and S (shift function) for 'S'.

If WEAK > 0, then non-elastic columns of the R-matrix are set to zero, when penetrabilities < WEAK.

KRM Change defined coupling order of R-matrix parameters. Best use default KRM=0.

EIGENS, MEIGS, VSEARCH, ECHAN, ENODES : if EIGENS, then:

MEIGS = maximum number of bound states to find,

VSEARCH = location of potential form to adjust so that channel ECHAN has ENODES nodes.

NPARAMETERS Read in this number of R-matrix parameter namelists after the &FRESCO namelist.

EOBS : if EOBS=T, treat input R-matrix widths as ‘observed’ values. Otherwise treat at ‘formal’ widths.

3.2.8 Trace variables

A value of 0 gives no trace, increasing values give progressively more printed output.
*Decrement*ed variables are decreased by 1 on each use.

CHANS ≥ 1 : Print the sets of coupled partial waves for each J,parity. Decrement.

LISTCC = 1 : Print coupling coefficients between these channels. Decrement.
= 1,2,... Print progressively more detail of couplings.

TRENEG ≥ 1 : Print all multipole potentials
 ≥ 3 : Print all monopole potentials as well

CDETR ≥ 1 : Print information on the solving of the coupled equations. (decrement).

SMATS ≥ 1 : Print absorbtion & reaction cross sections for successive partitions and excitations.
 ≥ 2 : Print elastic S-matrix elements (S_{el}). Also ‘punch’ these elastic S_{el} on output file 7, in format (2F15.10,L,J,JTOT) for S_{el} complex, L, J and JTOT. See WDISK below for description of these quantum numbers.
 ≥ 3 : Print all S-matrix elements for the ‘grazing partial waves’ defined by $0.05 < Re(S_{el}) < 0.95$
 ≥ 4 : Always print all the S-matrix elements.
 ≥ 5 : Print all S-matrix elements at each iteration of the coupled equations (or, if PADE > 0 , the Pade approximant)
 ≥ 6 : Print all actual S-matrix elements at each iteration (these may be divergent before Pade acceleration).

3.2.9 Output details

XSTABL, NLPL, WAVES, LAMPL, VEFF, KFUS, NFUS, WDISK, BPM, MELFIL, CDCC, TCFILE, tcfilename

XSTABL $\neq 0$: If XSTABL is non-zero, in file 16 punch output cross sections for all excitation levels in all partitions. A header line in FORMAT(5I6) gives partition IC, level pair IA, number of tensor ranks of analyzing powers $1 < KQ1PR < XSTABL$, number of angles NANGL, and NEARF. NEARF=1 for total cross section, 2 for far-side component, and 3 for near-side component.

Then follow NANGL print operations in FORMAT(1P, 6E12.4), repeating the FORMAT for each operation if KQ1PR is large, of THETA, elastic xs (mb), T_{10} , iT_{11} , T_{20} , T_{21} , T_{22} , iT_{30} , iT_{31} etc.

NLPL > 0 : print a ‘contour plot’ of the non-local kernels $K_{fi}(R_f, D_{fi})$. This is useful to determine if the parameters in the &Fresco namelist are adequate. Decrement.

WAVES ± 1 or ± 3 : print out wave function solutions of the coupled equations at the end of the iterations. (If WAVES <0 : print out the RATIO of the w/f to its asymptotic form $((G - iF) - S.(G + iF)).i/2$)
2 or 3 : print out the source terms at *each* iteration of the coupled equations.

LAMPL $\neq 0$: Print out (on Fortran file 36) the coefficients $A(m'M' : mM; L)$ for the Legendre coefficients in the scattering amplitude for the partition number abs(LAMPL), and print out the $f(m'M' : mM; \theta)$ for

each angle θ .

< 0 : only print out on file 37 the amplitudes f , not the A 's, for partition abs(LAMPL).

The phase convention here is that there is no Coulomb phase shift for $L = 0$ in the Coulomb scattering amplitude : factors such as $\exp i(\sigma_L - \sigma_0)$ appear in the A 's.

VEFF $\neq 0$: Calculate the 'coupled channels effective potential' found by averaging the 'trivially equivalent potential' over all the J, π sets, with weights of the elastic wave functions times the reaction cross section, in each set.

< 0 : Add this effective potential to optical potential of the elastic channel before printing.

$= -2$ or $+2$: Exclude partial waves with elastic S -matrix element $S_\ell < 0.1$ from the averaging sum.

The results show the real and imaginary parts for successive values of $J - L$, for the projectile only.

KFUS, NFUS If **KFUS** > 0 : Calculate 'core fusion' using the imaginary and scalar parts of potential number **KFUS** (i.e. &pot namelists with **TYPE** = 1 or 2, and **KP** = **KFUS**), also for the first **NFUS** inelastic channels.

WDISK = 1 : Print elastic wave functions on output file 17, FORMATTED

= 2 : Print all wave functions on output file 17, FORMATTED

= -1 : Print elastic wave functions on output file 17, UNFORMATTED

= -2 : Print all wave functions on output file 17, UNFORMATTED

The following data formats are used when **WDISK** > 0 :

line A: (I4, 2F8.4, F8.1, I3)

NR, H, ENLAB, JTOTAL, PARITY, MP, MT, ZP, ZT :

number radial points, step size, lab. energy, J, pi,
projectile and target masses and charges

line B: (2I4, 2F6.1, I4, F6.1, 2F15.10, f12.8)

IT, L, J, JTOT, LIN, JIN, SMAT (complex), ETA

where

IT = index to excited state pair, counts lines 7.

L = partial wave

J = L + projectile spin

JTOT = total spin = $J_{total} = J$ + target spin

LIN = incoming partial wave

JIN = incoming J value.

SMAT = S matrix element for this partial wave.

line C: (6E12.4) (psi(I), I=1, NR) wave function

line C is repeated until NR complex values given

NB: the first point $\psi(1) = 0$ always, as at $r=0$

Lines B & C are repeated for each channel, until $IT < 0$.

When **WDISK** < 0 , successive records contain the two real values of $\psi(I)$, starting IN THIS CASE, from $I=2$ (i.e. $r = h$).

BPM ≥ 1 : Calculate fusion cross sections in the Barrier Penetration Model using first the bare potential, and then the bare potential + the ‘weighted equivalent potential’ calculated when $VEFF \neq 0$.
 ≥ 2 : Print out L -distributions of the fusion cross section.

MELFIL $\neq 0$: Write real files 53 and 54 in the ‘mel’ and ‘spec’ format for use e.g. by R-matrix program sturmxx.
 $= \pm 2$: Write complex files 53 and 54 in the ‘mel’ and ‘spec’ format.
 < 0 : Write file 53 ‘mel’ in text format format.

CDCC = 1 : Print out the $f(m'M' : mM; \theta)$ for each angle θ on file 57 for partition PEL, after the following information for *uncoupled* bin states:

```

line Y: (i2) 1      (indicating CDCC=1 format below)
line Z: (A120)      HEADNG from Fresco input.
line A: (F10.4,3F8.4)  ENLAB,Bproj,H2SM,e^2,Btarg,inp, (Qval if inp=1)
                    lab energy,projectile binding energy, hbar^2/2.m, e^2,
                    target binding energy,inp, Qval if inp=1

line B: (7f8.4)      massp,masst,massc,massv,massr
                    masses: projectile,target,core,valence,residual
line C: (7f8.4)      Zp,Zt,Zc,Zv,Zr      charges
line D: (7A8)      namep,namet,namec,namev,namer names
line E: (7f8.1)      Jp,Jt,Jc,Jv,Jr      g.s. spins
line F: (7i8)      Pp,Pt,Pc,Pv,Pr      g.s. parities
    If inp=1, cards B-F (incl) have further #6 and #7 values
    for 'initial projectile' and 'initial target' too.

line G: (4I4)      NBINS,NKMAX,NEXB,NNJMAX no. CDCC bins, max NK,
                    no. excited states, max(2*Jex+1)
line H: (I4,2f8.4)  NANGL,THMIN,THINC      (cm angular range from \&FRESKO)
for each of the NBINS bins:
    line I: (i2,2f4.1,3f8.4,2i4)
        l,j,Emid,kmin,kmax,NK,KN,ISC
        l,j: quantum numbers (s==Jv)
        Emid: centre of bin with respect to continuum threshold
        kmin,kmax,NK: Min,max and number of k values in bin integral
        KN: original KN index for bin state
        ISC: normalisation used for bin
    for each IK=1,NK
        line J: (10f8.4) delta(IK): nuclear phase shift used in bin integral (radians)

for each excited state pair in the entrance partition: IA=1,NEXB::
    line K: (f4.1,i4,f8.4,i4) Jex,Parity,Eex,IBIN:
        Jex : spin of this projectile excited state (not including core spin)
        Parity: parity of this projectile state
        Eex: excitation energy of this state above g.s.
        IBIN: (first) bin defined for this excited state
    for each IANG=1,NANGL: read complex numbers:
    line L: (6E12.4): ((FAM(MEX,MP,IANG,IA),MEX=1,2*Jex(IA)+1),MP=1,2*Jp+1)

```

The phase convention for all CDCC values is that there is no Coulomb phase shift for $L = 0$ in the Coulomb scattering amplitude : factors such as $\exp i(\sigma_L - \sigma_0)$ appear in the A 's.t

Summary of bin normalisation factors for different ISC values:

ISC

= 2: $\exp(-i\delta(k))$
 = 4: $\sin(\delta(k))\exp(-i\delta(k))$
 = 12: $k\exp(-i\delta(k))$
 = 14: $k\sin(\delta(k))\exp(-i\delta(k))$

CDCC = 2 : Print out the $f(m'M' : mM; \theta)$ for each angle θ on file 57 for partition PEL, after the following information for *coupled bin states*:

```

line Y: (i2)  2      (indicating CDCC=2 format below)
line Z: (A120)  HEADNG from Fresco input.
line A: (F10.4,3F8.4)  ENLAB,Bproj,H2SM,e^2,Btarg,inp, Qval if inp=1
                        lab energy,projectile binding energy, hbar^2/2.m, e^2,
                        target binding energy,inp, Qval
line B: (7f8.4)      massp,masst,massc,massv,masrr
                        masses: projectile,target,core,valence,residual
line C: (7f8.4)      Zp,Zt,Zc,Zv,Zr      charges
line D: (7A8)      namep,namet,namec,namev,namer names
line E: (7f8.1)      Jp,Jt,Jc,Jv,Jr      g.s. spins
line F: (7i8)      Pp,Pt,Pc,Pv,Pr      g.s. parities
      If inp=1, cards B-F (incl) have further #6 and #7 values
      for 'initial projectile' and 'initial target' too.

line G: (5I4)      NBINS,NKMAX,NEXB,NNJMAX,NCHMAX
                        no. CDCC bins, max NK,
                        no. excited states, max(2*Jex+1), max nch
line H: (I4,2f8.4)  NANGL,THMIN,THINC      (cm angular range from \&FRESKO)
line I: (I4)      NCE      (number of excited states. NCE=0 for only gs).
for each of the ICE=1...NCE core excited states (card skipped if NCE=0)
lines J: (I4,2f8.4)  IPARCE, JCE, ECE      (parity -1,+1; spin; energy of excited states)
                        (for the gs: parity=Pc, spin=Jc, energy=0.0)

for each of the NBINS bins:
  card K: (f4.1,2I4,f4.1,3f8.4,2i4)
    Jex,Pex,nch,Emid,kmin,kmax,NK,KN,ISC
    Jex: overall spin [ (1 s)j, JCE(ICE); Jex>
    Pex: overall parity = parity(ICE) * (-1)**1
    nch: number of partial wave channels coupled to Jex/Pex.
    IL: incident channel (1<= IL <= nch)
    Emid: centre of bin with respect to continuum threshold
    kmin,kmax,NK: Min,max and number of k values in bin integral
    KN: original KN index for bin state
    ISC: normalisation used for bin
  For each partial wave c=1..nch
    line L: (i4,f4.1,i4)
      l,j,ICE: quantum numbers. Use s=Jv.
  for each IK=1,NK for k=kmin+(IK-1)*kinc where kinc = (kmax-kmin)/(NK-1)
    line M: (2f10.6) delta(IK),k: any nuclear phase shift used in bin integral (radians)
    line N: (10f10.6) S(:, :) the full scattering S matrix for nch channels
                        (always nch*nch, so closed channels included)

for each excited state pair in the entrance partition: IA=1,NEXB::
  line K: (f4.1,i4,f8.4,i4) Jex,Parity,Eex,IBIN:
    Jex : spin of this projectile excited state (including core spin)
    Pex: parity of this projectile state

```

```

      Eex:                excitation energy of this state above g.s. (including core energy)
      IBIN:                (first) bin defined for this excited state
for each IANG=1,NANGL: read complex numbers:
line L: (6E12.4): ((FAM(MEX,MP,IANG,IA),MEX=1,2*Jex(IA)+1),MP=1,2*Jp+1)

```

TCFILENAME : File to write HF data, default fort.5420.

TCFILE : Print in file TCFILNAME data needed for Hauser-Feshbach code YAHFC. These are penetration probabilities and all angular distributions.

DIST0 : If $\text{DIST0} \geq 0$, cut off charged-particle elastic scattering at below DIST0 degrees, for making GNDS or ENDF evaluation files. Default -1.

TMP : name of directory for temporary files: '/tmp' or '.' Default = /tmp/

GRACE, BOLDPLOT : for use after SFRESCOX searches.

If GRACE=T, print plotting metadata for xmgrace, not xmgr. Default=T.

BOLDPLOT: make width=2 lines in plots, not width=1, and all in monochrome. More suitable for publications.

3.3 &PARTITION and &STATES namelists

The &PARTITION namelists are repeated until an empty &PARTITION namelist is read in. After each non-empty ($\text{NEX} > 0$) &PARTITION, NEX instances of the &STATES namelist are required.

3.3.1 &PARTITION namelist

NAMEP, MASSP, ZP, and NAMET, MASST, ZT

NAMEP = name of projectile-like nuclide (string 8 characters max),

MASSP = mass of projectile (in units of UNITMASS),

ZP = charge of nuclide,

NAMET, MASST, ZT = equivalent properties of target-like nuclide.

NEX, QVAL, PWF

abs(NEX) = number of excited-state pairs in this partition.

If $\text{NEX} < 0$, then cross sections for this partition are not printed

QVAL = Q -value (in MeV) of this partition for $E_p + E_t = 0.0$. Only differences in Q -values between partitions are significant.

PWF = use asymptotic piecewise methods in this partition (default is True).

READSTATES, PRMAX, LPMAX, MIXPOT

READSTATES = read in specification of the NEX states from file number READSTATES.

PRMAX = maximum radius in this partition. Useful for R-matrix calculations, but redefines what 'local' interactions mean.

LPMAX = maximum partial wave L in this partition if the incident lab energy is below ELPMAX (given in the &Fresco namelist). Default=-1 (no limit),

MIXPOT governs the couplings that are generated (in this partition) between levels that have *different* KP potential indices. It does not affect any couplings when the KP index values are identical.

≤ 0 : no couplings (default)

= 1: couplings only to or from the first state in that partition

= 2: couplings only according to the KP value of the destination level

= 3: *all* couplings produced by either KP value.

3.3.2 &STATES namelists

Repeated IA = 1,...,NEX times before the next &PARTITION:

Jp, COPYp, BANDp, Ep, KKp, Tp, Jt, COPYt, BANDt, Et, KKt, Tt

For both projectile (p) and target (t), the following are read:

J = spin of the state (Jp and Jt, and similarly below).

BAND > 0 for positive parity state, and < 0 for negative parity. The value of BAND defines the rotational band : all states of the same BAND may be coupled by rotational mechanisms. Odd multipoles couple BAND values of opposite sign.

If COPY is positive, then this level is a copy of a previous level number (COPY < IA) in the same partition.

If COPYp < 0, then this level is an exchange copy of *the same IA numbered* level in the previous partition abs(COPY). That is, a copy with projectile and target nuclei exchanged.

If COPYt = -IC, then the target state is an exchanged projectile. If that state has spin zero, then only even partial waves are allowed.

COPYs are needed to define identical states of one nucleus in different pairings IA. Spectroscopic amplitudes (in &cfp namelists) should be defined for only the original 'copied' level.

E = excitation energy of state relative to ground state.

KK = K of rotation band containing this state. (If zero, use spin of first level in this band.)

T = isospin of this state (if zero, use INT((mass-2*charge)/2))

KP, FEXCH, IGNORE, INFAM, OUTFAM

KP = index of optical potential for this pair of excited states. If zero, use as default KP = IC, the number of this partition.

If FEXCH = T calculate and write out all cross sections and amplitudes for $180 - \theta$.

If IGNORE = T, ignore the convergence of this state pair (see IPS discussion above).

If INFAM $\neq 0$, read in complex amplitudes later to add to calculated scattering amplitudes
< 0, read in only one amplitude (non spin-flip) for each scattering angle, with others zero.

If OUTFAM $\neq 0$, write out complex amplitudes later to add to calculated scattering amplitudes.
< 0: write out only one amplitude for each angle, namely the average of diagonal amplitudes.

The output files are suitable to be reread later as input file

3.4 &POT potential namelists

Each potential is started by a &pot namelist with TYPE=0, defining the radii to be used and the diagonal Coulomb potential, followed by any number of further &pot namelists with TYPE>0, which cumulatively construct the components of the nuclear potential. The KP value in each namelist identifies the potential of which this is to be a component. The &pot namelists are repeated until KP< 0 indicates the last one to be considered, or an empty namelist is read in.

3.4.1 First kind of &pot namelist

KP, TYPE, p1, p2, p3, p4 with TYPE=0, or

KP, TYPE, AT, AP, RC, AC with TYPE=0 :

All the radii are multiplied by CC.

CC is updated by a TYPE=0 &pot namelist to $CC = AT^{1/3} + AP^{1/3}$ where $AT \equiv p1$ and $AP \equiv p2$,

CC is updated by TYPE>0 namelists, if $P0 > 0$, to $CC = P0^{1/3}$.

$RC \equiv p3$ = surface radius (when multiplied by CC) of charge distribution

$AC \equiv p4$ = diffuseness of charge distribution

The values of AT and AP affect *only* the conversion from r_0 radii to R radii in the definition of potentials in these namelists. If however you adopt the convention that AT is the target mass number and AP the projectile's, then FRESCOX can later display the fractional β values corresponding to given deformation lengths. (These are for information only, and do not affect the couplings.)

3.4.2 Second kind of &pot namelist: TYPE>0

KP, TYPE, IT, SHAPE, p1, p2, p3, p4, p5, p6, p0 or

KP, TYPE, IT, SHAPE, p(:) or

KP, TYPE, IT, SHAPE, V,Vr0,Va, W,Wr0,Wa

also: **LSHAPE, JL, XLVARY, ALVARY, DATAFILE**

An empty namelist ends reading potential parameters, as does a negative KP value in the last namelist.

For TYPE

0		Coulomb potential (see above)
1	SHAPE	Central potential, Volume
2	SHAPE	Central potential, Derivative (i.e. surface)
3	SHAPE	Spin-orbit for projectile
4	SHAPE	Spin-orbit for target
5	SHAPE	Tr tensor force for projectile
6	SHAPE	Tr tensor force for target
7	SHAPE	Tensor force between L and combined projectile+target spins (e.g. n-p tensor force)
8	SHAPE	Spin.spin force for target & projectile spins

9	SHAPE	Effective mass: reduction from unity
10	SHAPE	Deformed projectile (matrix elements from ROTOR)
11	SHAPE	Deformed target (matrix elements from ROTOR)
12	SHAPE	Projectile coupled by matrix elements read in.
13	SHAPE	Target coupled by matrix elements read in.
14	SHAPE	Projectile second-order coupled by matrix elements read in.
15	SHAPE	Target second-order coupled by matrix elements read in.
16	SHAPE	Target & projectile simultaneous second-order coupled by matrix elements read in.
17		Target & projectile all-order coupled by matrix elements read in.
18		Generate deformed spin-orbit forces from preceding TYPE=11 target deformations.
20	NUM	Super-soft N-N potl. of de Tourreil & Sprung (SSC(C))
21	NUM	User-supplied N-N potential via subroutine NNPOT.

TYPE indicates type of spin tensor or excitation coupling,

IT is 1 or 3 to make this component included only iteratively,

2 or 3 to NOT subtract this component in KIND=3,4 single-particle couplings,

'-' to NOT use this form to couple channels (only to make other forms, e.g. deformations).

NOUSE true ('T') to NOT USE this form to couple channels (only to make other forms, e.g. deformations),

SHAPE indicates the radial shape of the form factor,

P1-P3 & P4-P6 are parameters for the real and imaginary parts (respectively).

P0≡P7 can give a further parameter.

If TYPE < 0, add new potential numerically into previous potential, and use abs(TYPE) instead. This saves space, and is useful if adding components to be deformed by a following namelist with $\leq$ TYPE $\leq$ 13.

The SHAPEs for volume potentials (TYPE=1, 8 & 15) are,
with $R = P2 * CC$, $RH = (r - R)/P3$, and $E = \exp(-(r-R)/P3)$,
(replacing (P1,P2,P3) by (P4,P5,P6) throughout for the imaginary parts)

0: Woods-Saxon = $- P1 / (1 + 1/E)$

1: WS squared = $- P1 / (1 + 1/E)^2$.

2: Gaussian = $- P1 * \exp(-RH^2)$.

3: Yukawa = $- P1 * E / r$

4: Exponential = $- P1 * E$

5: Reid soft core for $T=0$, central part

6: Reid soft core for $T=1$, central part

7: Read Real

8: Read Imaginary

9: Read Complex

-1: Fourier-Bessel = $j_0(RH) = \sin(RH)/RH$

For SHAPEs 7-9, immediately read in free format from *Input File 4*, or from **datafile** if specified: First a comment line,

then (free format) NPOINTS, RSTEP, RFIRST,

followed by NPOINTS potential points for the shape in steps of RSTEP starting at $r=RFIRST$.

Rescale the real part by P1, the imaginary part by P2, and use radius = $P3 * CC$ for subsequent deformation

calculations.

If SHAPE = -7, -8, or -9, rewind file 4 or **datafile** before reading from it.

If **datafile** = continue, then continue reading the previous file (do not rewind or reopen).

The SHAPes for surface potentials (TYPE=2), with R, RH and E as before, are the first derivative forms (normalised to -1 when E=1, if reasonable):

- 0 : Woods-Saxon = - P1 * 4 * E / (1+E)²
- 1 : WS squared = - P1 * 8 * E*E / (1+E)³
- 2 : Gaussian = - P1 * 2 * exp(-RH*RH) * RH
- 3 : Yukawa = - P1 * E * (1 + r/P3) / (r*r)
- 4 : Exponential = - P1 * E
- 5 : Reid soft core for T=0, spin-orbit part $\times r$,
- 6 : Reid soft core for T=1, spin-orbit part $\times r$,
- 7,8,9 : Read from external file, as above.

The SHAPes for spin-orbit potentials (TYPE=3 or 4) are the following surface shapes, where $\text{CONLS} = \hbar^2 / (m_\pi^2 c^2) = 2.000$.

- 0 : Woods-Saxon = - CONLS * P1 * E / [(1+E)² * P3 * r]
- 1 : WS squared = - CONLS * P1 * E*E / (1+E)³ / (2*r*P3)
- 2 : Gaussian = - CONLS * P1 * exp(-RH*RH) * RH / (2*r*P3)
- 3 : Yukawa = - CONLS * P1 * E * (1 + r/P3) / (4*r³*P3)
- 4 : Exponential = - CONLS * P1 * E / (4*r*P3)
- 5 : Reid soft core for T=0, spin-orbit part
- 6 : Reid soft core for T=1, spin-orbit part
- 7,8,9 : Read from external file, as above.

To become interaction potentials, these form factors will be multiplied by $j(j+1) - l(l+1) - s(s+1) = 2\mathbf{l} \cdot \mathbf{s}$, where s = spin of projectile or target in each of its ground state and excited levels. There is no option here for state-dependent spin-orbit forces.

The SHAPes for tensor potentials (TYPE=5, 6 & 7), with R, RH and E as before, are the second derivative forms (normalised to unity when E=1, if reasonable):

- 0 : Woods-Saxon = P1 * 8 * E*(E-1) / (1+E)³
- 1 : WS squared = P1 * 4 * E*E / (1+E)³
- 2 : Gaussian = P1 * 2 * exp(-RH*RH) * RH
- 3 : Yukawa = P1 * E * (1/P3² + 2/r²) / r
- 4 : Exponential = P1 * E
- 5 : Reid soft core, 12 * tensor part
- 6 : Vol Gaussian = 12 * P1 * exp(-RH*RH)
- 7,8,9 : Read from external file, as above.

If SHAPE ≥ 30 , then use SHAPE-30, and define a J - or L -dependent potential by a factor defined using **JL, LSHAPE, XLVARY, ALVARY**

so that the potential is multiplied by a form factor depending LSHAPE as

=0 : Woods-Saxon = $1 / (1 + 1/E)$,
 =1 : WS squared = $1 / (1 + 1/E)^2$
 =2 : Gaussian = $\exp(-RH^2)$,
 where $X=L$ (for $JL='L'$), $X=J$ ($JL='J'$), $RH = (X-XLVARY)/ALVARY$, and $E=\exp(-RH)$.

If $SHAPE \geq 10$ and < 19 , then use $SHAPE-10$, and write out the resulting potential to *Output file 25* in `FORMAT(6E12.4)`: a form suitable for subsequent reading by a $SHAPE \geq 20$.

If $SHAPE \geq 20$ and < 30 , then a J_T/π -dependent potential is defined.
 For each subsequent coupled-channels set with total angular momentum J_T , a form factor is read into this potential from *Input file # SHAPE*, reading $M = RMATCH/HCM+1$ complex numbers in `FORMAT(6E12.4)` from the file as a random-access, with $(M-1)/3+1$ lines for each form factor.
 For each CC set, form factor number $int(J) + 1$ (for $SHAPE \geq 24$) or $1/2$ for $\pm$ parity (for $SHAPE < 24$) is read.

If $SHAPE = 40$, then a π -dependent potential is defined: use $KP = P1$ for $+$ parity CC sets, and $P2$ for $-$ parity.

If $SHAPE = 41$, then a L -dependent potential is defined: use $KP = P(L+1)$ for $L=0-5$, and $P0$ for $L \geq 6$.

If $SHAPE = 42$, then a J -dependent potential is defined: use $KP = P(J_i+1)$ for $J_i=0-5$, and $P0$ for $J_i \geq 6$, with $J_i = int(J)$.

If $SHAPE = 43$, then a L -parity-dependent potential is defined: use $KP = P1$ for L even, and $P2$ for L odd.
 There need only be one `&pot` namelist for $40 \leq SHAPE \leq 43$. The values of `TYPE` and `P2-P7` are ignored. No files are used. Choices are listed in `fort.48`

If $SHAPE = 45$, then an energy-dependent potential is defined, using linear interpolation of the potential forms. The nodes are $KP=P1$ at energy $P2$, $P3$ at energy $P4$, etc., up to $KP=P9$ at energy $P10$ (MeV).

Nucleon - nucleon Potentials

The SHAPes for the nucleon-nucleon SSC (C) potential (`TYPE = 8`) are taken from the subroutine `SSCC` in `FRESCOX`. The input `NUM` is the number of angular momentum components stored, using the order $^1S_0, ^3S_1, ^3S-^3D, ^3D_1, ^1P_1, ^3P_0, ^3P_1, ^3P_2, ^3P-^3F, ^3F_2, ^1D_2$, and 3D_2
 Note that `TYPEs 8 & 9` can only be used in `KIND = 1` (LS -coupled) single-particle states.

The SHAPes for the user's nucleon-nucleon potential (`TYPE = 9`) are taken from the subroutine `NNPOT`, with `NUM` the same meaning as above.

3.4.3 Deformations

The SHAPes for deformed potentials (`TYPE  $\geq 10$`) are constructed either by reading external forms ($SHAPE = 7, 8$ or 9) for each $P(k)$ non-zero (the input k -form being multiplied by $P(k)$), or (for $SHAPE \geq 10$) by deforming the potential on the *immediately preceding* input namelist. This preceding namelist must be of `TYPE=0` for Coulomb deformations, or `TYPE = 1` or 2 for nuclear deformations: deformation of non-central forces is not yet implemented.

For Nuclear deformations $P(k) = DEF(k)$, the deformation lengths (in fm.). ($P0 = DEF(0)$ for *inelastic* monopoles.)

For Coulomb deformations $P(k) = Mn(Ek)$, the intrinsic reduced matrix elements in units of $e.f m^k$ (P0 gives $Mn(0)$ for *inelastic* monopoles). These intrinsic matrix elements $Mn(Ek)$ use the reduced matrix elements $\langle I' || Ek || I \rangle$ given by the definitions of Alder & Winther (not by those of Brink & Satchler), by assuming a rotational model where K is a good quantum number:

$$Mn(Ek) = \langle I' || Ek || I \rangle / (\sqrt{2I+1} \langle IKk0 | I'K \rangle) . \quad (2)$$

Thus, for (possibly hypothetical) $0 \rightarrow k$ transitions, $Mn(Ek) = \langle k || Ek || 0 \rangle$.

$Mn(E2)$ is related to the intrinsic quadrupole moment in the rotational model by $Mn(Ek) = Q_0 \sqrt{5/16\pi}$. For a uniformly charged sphere of radius R , to first order in β_k we have

$$Mn(Ek) = \frac{3 Z \beta_k R^k}{4\pi} . \quad (3)$$

The $Mn(Ek)$ are thus of the same sign as β and Q_0 for all rotational bands.

TYPES 12 & 13, for detailed and model-independent specifications, directly use the reduced matrix element

$$M(Ek) = i^{I-I'+|I-I'|} \langle I' || Ek || I \rangle \quad (4)$$

$$= \pm \sqrt{(2I+1) B(Ek, I \rightarrow I')} , \quad (5)$$

where the sign convention has been chosen to make $M(Ek)$ real and symmetric under $I \leftrightarrow I'$ interchange. The diagonal reduced matrix element gives the spectroscopic quadrupole moment

$$Q_2 = \sqrt{16\pi/5} (2I+1)^{-1/2} \langle II20 | II \rangle \langle I || E2 || I \rangle . \quad (6)$$

The intrinsic and reduced matrix elements are therefore connected by

$$M(Ek) = Mn(Ek) (-1)^{[I-I'+|I-I'|]/2} \sqrt{2I+1} \langle IKk0 | I'K \rangle \quad (7)$$

in the rotational model where K is a good quantum number, for states of spin I, I' , projection K , charge Z , and radius R ($DEF(k) = \beta_k \times R$ for such states).

So $Mn(Ek)$ is the square root of $B(Ek, 0 \rightarrow k)$ with some sign within $K = 0$ bands. $Mn(Ek)$ always has the same sign as β .

*In Other Cases, Do **Not** Assume That All Transitions Have The Same $M(Ek)$ Values. Thus $Mn(Ek)$ Are The Same For All Pure Rotor Transitions, In Contrast To $M(Ek)$.* In order to guide the user here, effective β values are calculated from the given $Mn(Ek)$ or $M(Ek)$, for each possible transition. *From version FRV onward, $Mn(Ek)$ is used in place of $M(Ek)$ for TYPE 10 & 11 potentials.*

3.4.4 Pairwise couplings

For TYPEs 12–17, namelists **&step** are read to determine the details of the required couplings between each pair of excited states and of the reorientation terms for each individual state.

&step IB,I A, k, STR /

to put in a coupling *to* state IB *from* state IA, of multipolarity k and strength STR. These are terminated by an empty **&step** namelist, or by IB<0 indicating the last entry.

For each k used here, P(k) must have been non-zero, and the required coupling has the same form factor but with relative strength STR / P(k). That is, STR is the reduced matrix element $\langle IB|M(Ek)|IA \rangle = M(Ek, IA \rightarrow IB)$ for Coulomb transitions (**NOT** Mn(Ek)), and is the ‘*reduced* deformation length’ RDEF(k) for nuclear transitions.

This ‘reduced deformation length’ in the rotational model is

$$RDEF(k, I \rightarrow I') = DEF(k)(-1)^{[I-I'+|I-I'|]/2} \sqrt{2I+1} \langle IKk0|I'K \rangle \quad (8)$$

$$= M(Ek) * 4\pi / [3ZR^{k-1}] \quad (9)$$

but may in general be varied independently of DEF(k) and M(Ek). Both these nuclear and Coulomb interactions are thus designed to give channel couplings which are *independent* of the projection K values read in the **&state** namelists. This of course requires making RDEF(k) and M(Ek) dependent on the spins of the excited states, not on any simple model. Note: monopole potentials are included automatically, even if not listed in a **&step** namelist.

The SHAPes for deformed potentials (TYPE=10 to 13) are constructed by:

SHAPE = 0 to 6 : not used

= 7,8,9 : Read from external file, as above for TYPE=1. The multipoles are then multiplied by P(k) (by 1.0 for k=0), and should end up as scaled to replace the form factors calculated below:

= 10 : Coulomb multipoles have the usual simple form for a deformed charged sphere to produce M(Ek) = P(k), varying at large distances as

$$F(r) = M(Ek) e^2 \frac{\sqrt{4\pi}}{(2k+1)} r^{-k-1} \quad (10)$$

where $e^2 = 1.440$, and where this form factor will later be multiplied by the charge number of the opposing nucleus.

= 10 : Nuclear multipoles are the deformation length DEF(k) = P(k) times the derivative of the undeformed potential times $-1/\sqrt{4\pi}$:

$$F(r) = -DEF(k) \frac{1}{\sqrt{4\pi}} \frac{dU(r)}{dr} \quad (11)$$

The shape is the same for all multipoles $k > 0$.

= 11: Nuclear multipoles calculated by numerically deforming the radii of the previous potential, and projecting by Gaussian quadrature onto the required multipoles (those $k > 0$ for which P(k) is non-zero). For small deformation lengths, these SHAPes are the same as for SHAPE = 10.

= 12: Same as SHAPE = 11, except that the monopole $k = 0$ part of the potential is also recalculated by quadrature. A first order correction is made to the radius to improve volume conservation.

= 13: Same as SHAPE = 12, except that **no** first order correction is made to the radius to improve volume conservation.

For TYPEs 14–17, **&step** namelists are again read to determine the details of the required couplings between each pair of excited states and of the reorientation terms for each individual state. Only nuclear couplings are

yet implemented. Specify values **IB, I A, k, STR** to put in a coupling *to* state IB *from* state IA, of multipolarity |k| and strength STR. For TYPEs=14–17, the IB and IA can reference any combination of projectile and target states, but k>0 for projectile and k<0 for target states. STR refers to the RDEF(k) measure above; the factor $(-1)^{[I-I'+|I-I'|]/2}$ must be included.

For TYPE=17, couplings, to all orders in RDEF, are calculated between all pairs of projectile and target states using the matrix eigensolution method of Kermode and Rowley, Phys. Rev. **C48** (1993) 2326. The SHAPE and P0-P6 parameters on the TYPE=17 &pot namelist are not used.

For TYPE=18 generate deformed spin-orbit couplings from preceding TYPE=11 target deformations. The TYPE=11 namelist and the used central potential themselves should have nouse=T to remove those potentials themselves being included. Rescale the real part by P1, the imaginary part by P2.

3.5 &OVERLAP namelists for one- and two-particle form factors

Read in a namelist for each desired particle-nucleus form factor, then an empty namelist.

KN1, KN2, IC1, IC2, IN

Each form factor is indexed by a number KN, which may refer to a single form, or for two-nucleon form factors will refer to a range of forms KN1 - KN2 (one for each distance and angular momentum between the two nucleons).

Each form factor is the binding of one or two particles to a specified nucleus, where the composite system is another specified nucleus.

To specify the core and composite nuclei, their partition numbers IC1 and IC2 are required (either order), with $|IN|=1$ for projectile and $|IN|=2$ for target nuclei.

The mass of the particle is the strict difference of the masses of the core and composite nuclei, except that if $IN < 0$, then a relativistic correction is made for effect of the relative Q-values on the mass of the composite nucleus, when extracting by differences the mass of the bound fragment.

KIND the kind of coupling order (0 to 4 for one-particle states, and 6 to 9 for two-particle states) :

- = 0 for (LN, SN) JN couplings ← use for typical transfers
- = 1 for $|L_n, (SN, J_{core})S; J_{com}\rangle$ (IA & IB must be given) *
- = 2 for eigenstate in deformed potential (fixed SN, K, Parity) *
- = 3 for $|(LN, SN)J_n, J_{core}; J_{com}\rangle$ (IA & IB must be given)
- = 4 for form of leg of the Dalitz-Thacker Triton
- = 5 (not used)
- = 6 for $|L_{nn}, (\ell, S_{12})j_{12}; J_{12}\rangle$ with isospin $(.5, .5)T$
- = 7 for $|(L_{nn}, \ell)L_t, (S_{12}, J_{core})S_t; J_{com}\rangle$ & $|(.5, .5)T, T_{core}; T_{com}\rangle$ *
- = 8 (not used)
- = 9 for $(L_{nn}, (\ell, S_{12})j_{12})J_{12}, J_{core}; J_{com}\rangle$ & $(.5, .5)T, T_{core}; T_{com}$

where

J_{core} = spin of core nucleus (state Ia if given)

J_{com} = spin of composite nucleus (state IB if given)

K_{core} = projection K of core nucleus (state IA)

K_{com} = projection K of composite nucleus (state IB)

T_{core} = isospin of core nucleus (state IA)

T_{com} = isospin of composite nucleus (state IB)

and * signifies that transfers using these KINDs are not yet implemented.

CH1, NN, L, LMAX, SN, IA, JN, IB

CH1 = single-character identifier to distinguish clusters of nucleons of different structures that are not further described but should not be confused with each other. Use A-M for positive parity clusters, and N-Z for negative parities.

NN = number of nodes (include the origin, but not infinity, so $NN > 0$)

L = LN = angular momentum of bound cluster relative to the core

LMAX = maximum value of L in states in deformed potential,

SN = intrinsic spin of bound nucleon (one-particle states)
= total angular momentum (L+S) of bound cluster (KINDs 6 & 9)
= combined cluster and core intrinsic spins (KIND 7)

IA = index (within core partition) of excited state of core, or zero if to be specified later.

JN = vector sum LN + SN,
but for KIND=1, JN = SN + J_{core} (i.e. S in LS coupling)
and for KIND=7, JN = $L_{nn} + \ell$ (i.e. L in LS coupling)

IB = index (within composite partition) of excited state of composite, or zero if to be specified later.

KBPOT, KRPOT, BE, ISC, NK, ER

KBPOT = index KP of potential in which to bind this state

KRPOT = index KP' of potential with which to multiply this states' wave function for transfer interactions. If zero, use KBPOT. If the binding potential was adjusted for a specific binding energy, then this adjusted potential (not the original) is used for transfers.

BE = Binding Energy (positive for bound states, negative for continuum bins)

BOUND STATES:

ISC = 0 to vary the binding energy for fixed potential,

> 0 to vary the TYPE = ISC component of the potential KBPOT by a scaling factor to give binding energy BE.

< 0 to vary the TYPE = ISC component of the potential KBPOT, as above, but **also** to permanently rescale all the varied potential components. This affects all later bound and scattering states using the potential KBPOT.

CONTINUUM BINS:

ISC (default value 2)

= -2: no weighting or normalisation,

= 1, 2 : weight wave functions by $\exp(-i\delta(k))$, so they are real, before integrating over bin width.

= 3, 4 : weight wave functions by $T(k)^*$ (useful for resonances)

= -1, 1, 3 : normalise wave functions to unity (by usual square norm). **(This option is not recommended, for physics reasons!)**

≥ 10 : use additional k factor in the weighting function, with $\text{mod}(\text{ISC}, 10)$ for above choices. Recommended for low-energy bins.

The values $\text{ISC} > 0$ give real-valued bins for single-channel states, but not, in general, for coupled-channels bins. In that case, Fresco stores *all* bound and continuum states as complex functions. KIND=3 and 4 forward and reverse couplings are both calculated explicitly, but not KIND=7 transfer couplings. For transfers, the reverse couplings must be put in explicitly.

Warning: imaginary parts of bins give imaginary parts of long-range Coulomb couplings that are *ignored* between `abs(RMATCH)` and `RASYM` in the `CRCWFN` calculations (the values in the `&Fresco` namelist).

In general, $\text{ISC}=2$ is recommended (or $\text{ISC}=4$ for resonances), since then the coupled-channels bins will be nearly real. With $\text{ISC}=-2$ (no phase weighting) there will be different results because of different weighting within the bin. Note that all channels within a bin have the same scalar weighting factor.

IPC, NFL

IPC = print control for further information:

IPC	Print Iterations	Print Final Result	Print W/F
0	no	no	no
1	no	yes	yes
2	no	yes	no
3	yes	yes	yes
4	yes	yes	no

For bin states, read this table with ‘iterations’ replaced by ‘intermediate phase shifts’.

NFL < 0 : to write wave-function $u(R)/R$ and potential overlap $V(R)u(R)/R$ to file number $\text{abs}(\text{NFL})$.

> 0 : to read a previously-written wave function from file number NFL.

These wave function files contain a comment line, then (free format) NPOINTS, RSTEP, RFIRST, followed by NPOINTS wf points in steps of RSTEP starting at $r=\text{RFIRST}$, and then NPOINTS for the vertex function (potential*wf). The file numbers NFL should be in the range 20–33 (see section 7).

NAM, AMPL

If IA and IB are both non-zero, then there is enough information to set up the spectroscopic amplitude now, to $\sqrt{NAM} \times \text{AMPL}$. See discussion for the &cfp namelists, for further information about these amplitudes.

Note: If this is a form factor with mixed core levels, then AMPL should be specified *here* rather than on a &cfp namelist (as &cfp namelists require an IA specification, and such a state would have multiple IA assignments).

If NAM = -1, then use AMPL for the mass of the bound particle, independent of the MASSES in the &partition namelists. (If NAM ≥ 0, then the default particle mass is the difference of the MASSES of the projectiles (IN=1) or targets (IN=2) for partitions IC1 and IC2).

If NAM < -1, then use AMPL instead of ERANGE for the range ER of the energies of the upper and lower boundaries of the continuum bins, and use at least $\text{gr NK}=5 \times |NAM|$ integration steps over this range. (These ER and NK override the input values.)

DM Set particle mass specifically, rather than using mass differences.

NK, ER, E Specific values for this continuum bin.

RSMIN, RSMAX If specified, set the overlap form factors to zero outside the interval [RSMIN,RSMAX]

NLAG, PHASE, AUTOWID Parameters for PLUTO with bound or resonant states.

3.5.1 One-particle KINDs

For KIND = 0: (LN,SN) JN couplings.

One form factor with LN,SN, & JN as read in.

LMAX is not used.

IA & IB are used only if NAM & AMPL are non-zero, to specify spectroscopic amplitudes.

For KIND = 1: $L_n, (SN, J_{core})S; J_{com}$ i.e. LS coupling.

Coupled form factors with sum over L_n and S .

SN as read in, the intrinsic spin of the bound particle.

IA giving J_{core} , spin of core nucleus.

IB giving J_{com} , spin of composite nucleus.

LMAX is maximum L_n in summation.

JN is maximum S in summation.

NN & L restrict the number of radial nodes of one component wave function in the coupled set :

NN gives the number of radial nodes of the last partial wave of angular momentum $L_n = L$ input.

For KIND = 2: eigenstates in a deformed potential, for fixed parity and K projection :

Coupled form factors $(l_n, SN)JN, K$, with sum over L_n & J_n .

SN, LMAX, NN & L are as for KIND = 1 :

SN as read in, the intrinsic spin of the bound particle.

LMAX is maximum L_n in summation.

NN & L restrict the number of radial nodes of one component wave function in the coupled set.

IA is a core state with correct K_{core} projection quantum number.

IB is a composite state with correct K_{com} projection number, so $K = K_{com} - K_{core}$

JN is the maximum J_n in the summation.

For KIND = 3: sum over coupled core and $(ls)j$ particle states :

Coupled form factors $(L_n, SN)J_n, J_{core}; J_{com}$ (summing over L_n, J_n & J_{core}).

NN is the required number of radial nodes for the component wave function with core state IA and partial wave $L_n = L$.

LMAX is maximum L_n in summation.

SN as read in, the intrinsic spin of the bound particle.

All core states are included that can be coupled to form J_{com} , using a deformed binding potential. *Note that such a deformed potential must be TYPE = 11, whether the projectile or target is deformed: not 10 or 12 or 13.*

JN is the maximum J_n in the summation (single particle $l + s$)

IB gives J_{com} (fixed) : spin of composite nucleus.

BE is the single-particle binding energy for core state IA.

3.5.2 Two-particle KINDs

For KIND ≥ 6 : two-particle bound states are constructed out of sums of pairs of previously-defined one-particle states, and input parameters NN through to BE in the &overlap namelist are given new meanings:

NN → NPAIRS, the number of pair-products to be summed

L → ℓ_{min} , minimum orbital angular momentum ℓ .

LMAX → ℓ_{max} , maximum orbital angular momentum ℓ .

SN → S_{min} , minimum sum S_{12} of the two nucleons' intrinsic spins ($S_{max} = 1.0$ always).

IA, IB give core and composite states, as before.

JN → J_{12} , total angular momentum of the two-particle state outside J_{core} .

KBOT → T, total Isospin of the 2-nucleon state (0 or 1)s (used to enforce $\ell + S_{12} + T = \text{odd}$)

KRPOT → KNZR, the KN index to a single-particle state $u_{12}(r)$ of KIND 0 or 1, giving the N-N relative motion in the other participating nucleus (usually in the light ion).

If $KNZR > 0$, then just the overlap $u(R) = \langle u_{12}(r) | U(r, R) \rangle$ is produced, suitable for zero-range two-nucleon transfer calculations.

BE → EPS, the threshold percentage to define components with square norms sufficiently small to be omitted in the final two-nucleon state.

If $ISC \leq 0$, use Gaussian quadrature grid, in blocks of 6th-order positions, for the N-N distance RMIN to RNN (&Fresco namelist).

If $ISC \geq 1$, use uniform grid for the N-N distances. Not so accurate.

If $ISC \neq 0$, print out numerical values of resulting two-nucleon wavefunction $U(r, R)$.

IPC controls the details printed (along with ISC as just above):

≥ 0 : one-line summary of $U(r, R)$ form factor for each r .

≥ 1 : overall norm and rms radius of total NN state.

≥ 2 : contour plot of the L, ℓ components included.

≥ 3 : contour plot of the $U(r, R) \times$ interaction potential

$NFL < 0$: to write two-nucleon wave-function $U(r, R)$ to file number $\text{abs}(NFL)$.

> 0 : to read a previously-written wave function from file number NFL . The values of NPAIRS, ℓ_{min} , ℓ_{max} , S_{min} of the *present* run are ignored. Thus NPAIRS can be set to zero.

3.5.3 Namelist &TWONT

This namelist gives the details of the single-particle wave functions and their amplitudes in the sum of pairs.

NT(1:4, :), COEF(:)

The sum over I of $\text{COEF}(I) |(l_1, s_1)j_1, (l_2, s_2)j_2; J_{12}, T\rangle$,
for $(l_1, s_1)j_1$ given by state $u_1(r_1)$ of $KN1 = NT(1, I) > 0$,
and $(l_2, s_2)j_2$ given by state $u_2(r_2)$ of $KN2 = NT(2, I) > 0$,
is then transformed into the required $KIND = 6$ format.

The (r_1, r_2) coordinates become (r, R) coordinates, where r = distance between the two nucleons (angular spin ℓ), and R = distance from the core to their centre of mass (corresponding angular momentum is L_{cm}).

If $NT(3, I) > 0$, then the wave functions $u_1(r_1) \times u_2(r_2)$ are further multiplied by $u_3(r)$ (of $KN3 = NT(3, I)$) before coordinate transformations. Only the radial shape of $KN3$ is used, not any angular momentum numbers.

If $NT(1,I) = -1$, then an external form factor is read in from Fortran file number $NT(2,I)$, and processed using the subroutine EXTERN1. At present, this routine is written to read triton wave functions from the Grenoble Faddeev calculations, only reading wave functions, *not* the potential $\times$ wavefunction (so for e.g. stripping, only use prior interactions).

If $NT(1,I) = -2$, then an external form factor is read in from Fortran file number $NT(2,I)$, and processed using the subroutine EXTERN2. At present, this routine is written to read 3-body wave functions from HH calculations, reading wave functions, *as well as* the potential $\times$ wavefunction. The 'vrr' file format is assumed, and the breakup (third) vertex function is ignored.

If $NT(3,I) < 0$, then the I 'th component of the pair summation is simply the product of cluster wave functions $u_1(r)u_2(R)$ where u_2 may be $KIND = 0$ or 1 .

$NT(4,I)$ is not used in this version of FRESCOX.

For $KIND = 6$: construct components of the form $L_{nn}, ((\ell, (s_1, s_2)S_{12})j_{12}; J_{12}, T)$ for all different L_{nn}, ℓ, S_{12} & j_{12} values permitted within the limits set by $\ell_{min}, \ell_{max}, S_{min}$, and S_{max} (J_{12} and T are fixed) by summing over pairs of single-particle wave functions.

For all KINDs, the printout also lists:

DZ = derived charge of the bound particle (always positive).

DM = derived mass of the bound particle

K = the wave number of the bound state asymptotically

NORM=overall square norm of this bound state. The wave functions of the single-particle bound states are always normalised to unity.

RMS= root-mean-square radius of this bound state

D_0 = zero-range stripping strength for transfers from this state

D = asymptotic stripping strength, as used e.g. in sub-Coulomb transfers

3.6 &COUPLING namelists

Now read in namelists for each kind of coupling between partitions and/or between excited states, ending with a namelist which is empty or has ICTO=0. Note that if deformed potentials were given as channel optical potentials then there will already be some couplings between excited states.

ICTO, ICFROM ,KIND, IP1,IP2,IP3, P1,P2, IP4,IP5

The coupling is from all the states in partition ICFROM to all the states in partition ICTO.

Couplings in the reverse direction are also included unless $ICTO < 0$, except for KINDs 1 & 2 where finer control is allowed.

JMAX, RMX Couplings are only active for $J < JMAX$ and Radius $< RMX$, (if $JMAX=0$, use JTMAX, and if $RMX=0$, use RMATCH).

KIND The couplings can be of 10 KINDs:

- = 1 : general spin transfer for projectile/transfer couplings
- = 2 : electromagnetic one-photon couplings
- = 3 : single-particle excitation of the projectile
- = 4 : single-particle excitation of the target
- = 5 : zero-range or LEA transfer with strength P1 and finite range radius P2 (in fm.)
- = 6 : LEA transfer using D_0 & D from bound states
- = 7 : finite-range transfer
- = 8 : non-orthogonality correction to a KIND 5,6,7 transfer
- = 9 : general partial-wave couplings
- = 10 : (spare)
- = 11 : Projectile-valence non-orthogonality
- = 12 : Target-valence non-orthogonality (not implemented)

After &coupling namelists for KINDs 2,3,4,5,6,7 (& 8 if IP2>0), spectroscopic amplitudes are read in by means of &cfp namelists. These coupling types are those for which ICTO & ICFROM are different, one being a core partition and the other a composite nucleus. One table is used for all the amplitudes: it is indexed (besides the partition numbers) by IN,IB,IA & KN (see glossary), and stores a real number A.

The table is filled cumulatively, starting perhaps with &overlap namelists, so amplitudes need only be entered after the namelist for the first KIND of coupling in which they appear. (So if all the amplitudes are already entered for a certain KIND of coupling, then the following namelist will be blank, to indicate no more &cfp namelists are to be read).

For KIND = 1: General Spin Transfer (with local or non-local external form factors on FILE 'INFILE' [default 4]) See section 8 for more details.

IP1

- = 0 : local form factor
- 1 : non-local (two-dimensional) form factor
- 2 : derivative local form factor
- 3 : derivative non-local (two-dimensional) form factor (not implemented)

IP2

- = 0 : read in real values (only if IP3 $\geq$ 0)
- 1 : read in imaginary values (only if IP3 $\geq$ 0)
- 2 : read in complex values (only if IP3 $\geq$ 0)

IP3

- = 3 : read data from JLM folding program jlmP, for projectile couplings only
- 2 : read data from JLM folding program jlmP, for target couplings only
- 1 : read data from charge exchange program CHEX2, with appropriate scaling.
- 0 : no jlmP or CHEX2 scaling factors, only P1, P2 and FSCALE
- 1 : write out typical non-local grid (R, R') coordinates
- 2 : calculate non-local grid (R, R') coordinates, and call subroutine FFNL to calculate form factor.
- 3 : calculate non-local grid (R, R') coordinates, call subroutine FNLSET to calculate an initial form factor, and the call FNLCC for each pair of coupled partial waves, for L-dependent factors.
- 4 : call subroutine FNLREAD to read in non-local form factor, and multiplying it by the Racah-algebra factor in section 8 (except for the $1/\sqrt{4\pi}$ factor) for given IB, IA final and initial excited states. The

order of form factors in the file must agree with the order called by a double loop of channels, with the 'to' channel number varying more rapidly.

-5 : call subroutine FNLREAD to read in non-local form factor for given IB, IA final and initial excited states. The order of form factors in the file must agree with the order called by a double loop of channels, with the 'to' channel number varying more rapidly.

P1,P2 : scaling for the real and imaginary parts (respectively).

If $IP3 \geq 0$, read FROM FILE 'INFILE' the following lines (until a blank line):

```
Format 16.2: I4, 3F8.4,           I4, 2F4.0, 4I4,           4A8
              NP ,HNP,RFS,FSCALE, LTR, PTR,TTR, IB,IA, LOP,DIR, COMMENT
for NP = number of radial points
  HNP = step size
  RFS = radius of first point
  FSCALE = scaling factor to be applied
  LTR = L-transfer
  PTR = projectile spin transfer
  TTR = target spin transfer
  IB = excited-state pair fed by coupling
  IA = excited-state pair feeding the coupling
  LOP = 0 for angular momentum transfers, 1 for ang-mom. operator itself
  DIR = 0 for radial scalar interaction, 1 for radial derivative
  COMMENT = comment for display in printout.
```

If LOP, DIR cannot be read without error, they will be assumed both to be 0.

Then free format for the radial shapes, repeatedly until sufficient data is read.

If $IP1 = 0$, read local form factor from $i=1$ to N

1, read successively for $j=1, NLO$ the non-local form factors $FNL(i,j)$, $i=1$ to NP [$NLO = RNL / \max(HNL, HCM)$]

When $IP3=1$ for CHEX2 input, the RFS and its F8.4 is omitted from Card 16.2, with default value $RFS=HNP$.

For KIND = 2: Electromagnetic one-photon couplings (for $E\lambda$ and $M\lambda$ processes).

ICTO is the gamma partition and ICFROM the particle partition. The photon must be on the 'projectile' side, and the bound state between target states.

$IP1 = \lambda$: The multipolarity of the radiation. If $\lambda > 0$, include all multipoles $1, \dots, \lambda$ permitted by parity, whereas if $IP1 < 0$, include only the multipole $\lambda = \text{abs}(IP1)$.

$IP2$

= 0,4 : Calculates both electric and magnetic couplings (reads g -factors from P1&P2)

1,5 : for electric only

2 : for magnetic only (reads g -factors from P1&P2)

4,5 : include also Siegert remnant for electric transitions

IP3: *not implemented yet*

P1 = projectile g -factor

P2 = target g -factor

IP4

= 0 : direct capture only

1 : semi-direct capture only

2 : direct + semi-direct capture mechanisms.

For KIND = 3 or 4: Single-particle excitations of the projectile (3) or target (4)

For these KINDs, ICTO is the partition of the nuclei being excited, and ICFROM is used to indicate the *core* partition if the single particle were removed. No couplings are generated to or from the ICFROM partition, only within ICTO partition.

IP1 = Q : The multipole order of the deforming potential due to the colliding nucleus. If $Q > 0$, include all multipoles 0,1,..., Q permitted by parity, whereas if $IP1 < 0$, include only the multipole $Q = \text{abs}(IP1)$.

IP2

= 0 : Coulomb & nuclear (complex)

= 1 : nuclear (complex) only

= 2 : Coulomb only

IP3

= 0 or 10 : include all re-orientation terms

= 1 or 11 : no re-orientation terms for $Q > 0$

= 2 or 12 : ONLY re-orientation terms

= 3 or 13 : include only couplings to and from the ground state, but NOT gs reorientation,

= 4 or 14 : include diagonal couplings, and couplings to and from the ground state

= 5 or 15 : include diagonal couplings, and couplings to and from any bound state

≥ 10 : read namelist **&scale** for complex factors [QSCALE(Q), $Q=\max(0,-IP1);\text{abs}(IP1)$] to scale the folded form factors for multipoles Q .

IP4 = Q_{max} , the max deformed core potential multipole

IP5 = Λ_{max} , the new multipole order for formfactor reduction.

For these core-excitation (XCDCC) options, the code needs to be compiled with the -Dcorex option.

P1 = FLOAT(potential KP index for fragment - target interaction)

P2 = FLOAT(potential KP index for core - target interaction)

(only SCALAR parts of the potentials P1 and P2 are used).

For KIND = 5 or 6: Zero-range & LEA transfers

IP1 and IP2 not used.

For KIND=5,

P1 = D_0 - ZR coupling constant

P2 = FNRNG - Effective finite-range parameter for use in LEA (in units of fm.)

For KIND=6, use D_0 and $\text{FNRNG} = \sqrt{(D/D_0 - 1)/k^2}$. from the projectile bound states. With unbound states, or if IP3 = 1, use D_0 from state, but FNRNG = P2 from this namelist.

In both KINDs, read in &cfp namelists, noting that for KIND = 5 spectroscopic factors for the projectile are not needed, and are ignored.

Users of these interactions kinds should also pay attention to the parameter INH in the &Fresco namelist.

For KIND = 7: Finite-range transfers.

IP1 =

0,-2 : POST interaction

1,-1 : PRIOR interaction (N.B. meaning of IP1 thus depends on ICTO & ICFROM!!!)

≤ -1 : Use θ quadrature from $\theta = \pi$ down to $\theta = 0$. (Useful for finite-range knock-on with light projectiles)

≤ -3 : VCORE interaction: Use ONLY the core-core interaction potential (Useful for finite-range knock-on with light projectiles)

2 : Surface transfer operator, on surface if *final* bound state $r' = P1$. IP2 and IP3 ignored.

3 : Surface transfer operator, on surface if *initial* bound state $r = P1$. (Not yet implemented)

4 : Surface transfer operator calculated as PRIOR-POST. Need also to set RSMAX for *final* bound state to the desired surface radius. (Implemented, but not yet correctly).

5 : Surface transfer operator calculated as POST-PRIOR. Need also to set RSMAX for *initial* bound state to the desired surface radius. (Not yet implemented).

IP2 =

0 : no remnant

1 : full real remnant

-1 : full complex remnant

2 : “non-orthogonality remnant” - this works by inserting a KIND = 8 coupling namelist after this coupling, before any subsequent couplings. This does not affect the one-step amplitudes, and is only useful if another transfer step follows this coupling.

-2 : “non-orthogonality remnant” and full complex remnant

IP3 = KPCORE : the number KP of the potential to use between the two cores, in the remnant part of the interaction potential.

If IP3=0, use as KPCORE the optical potential given for the first pair of excited states in the partition of projectile core. (this uses the observation that optical potentials tend to depend more on the projectile than the target, and must clearly be re-examined if the projectile is heavier than the target.)

P1,P2 : not used.

For KIND = 8: Non-orthogonality supplement appropriate to a previous KIND 5,6 or 7 interaction.

IP1 =

0 : post

1 : prior (N.B. IP1 should be the same as the previous interaction!)

IP2

> 0 : read in spectroscopic factors as &cfp namelists (Only useful if you did NOT have a previous KIND 5, 6 or s7 interaction, which would have needed the amplitudes then.)

$= 0$: no &cfp namelists to read.

Note that you should *either* use KIND=7, |IP2|=2, *or* use KIND=7, |IP2|=1 and a KIND=8. If you have KIND=7, |IP2|=2 and a KIND=8 namelist, then this is double counting.

For KIND = 9: General Partial-Wave Couplings (local or non-local external interactions from FILE 'INFILE' [default 4])

IP1

= 0 : local form factor

1 : non-local (two-dimensional) form factor

IP2

= 0 : read in real values

1 : read in imaginary values

2 : read in complex values

P1,P2 : scaling for the real and imaginary parts (respectively).

For IP1=0 (local), read from file 'INFILE' the following lines (until a blank line):

```
Line (free format)          NP, sHNP, RFS
      for NP = number of radial points
      HNP = step size
      RFS = radius of first point
```

For IP1=1 (nonlocal), read from file 'INFILE' the following lines (until a blank line):

```
Line (free format)          NLLI, RINTPI, NLOI, HNLI, NLCI
      for NLLI = number of radial points in R
      RINTPI = step size in R
      NLOI = number of radial points in D=R'-R
      HNLI= step size in D
      NLCI=offset index for D=0
```

Lines:

```
F8.1,I4,          2I4,f6.1,          2I4,F6.1          L2
JTOTAL,PARITY,  IB,LVAL1,JVAL1,IA,LVAL2,JVAL2,REV
```

each with following potential data of Cards 16.6 or 16.7. These are the couplings from partial wave *to* (LVAL1,JVAL1) for excited state pair IB *from* partial wave (LVAL2,JVAL2) for excited state pair IA, in coupled channels set JTOTAL of parity PARITY (+1 or -1). The reverse coupling is also included if REV is true.

If IB or IA is zero, match any state. If JVALi < 0, then match any JVALi. If PARITY=0, match any parity. If JTOTAL < 0, match any JTOTAL. If LVALi < 0, then match any LVALi. If LVAL1=LVAL2 < 0, then require partial wave coupling to be diagonal for equal LVAL=-LVALi-1.

For IP1=0 (local),

```
read (free format) NP values
```

For IP1=1 (nonlocal), read NLLI loops of:

read NLOI values (free format).

Go back to reading the above input files (though subsequent values in those files are ignored).

For KIND = 11: Projectile-valence non-orthogonality Construct non-orthogonality overlap, for use in R-matrix or Lagrange-mesh methods, for the case when a projectile p scattering on a target consisting of {core c + valence v } can be rearranged in another partition to be projectile' v scattering on a target' consisting of {core c + valence p }.

ICTO, ICFROM are the two partitions with p and v as projectiles (either order).

IP1 = ICORE is the partition in which the target is the bare core c .

Spectroscopic amplitudes need to have been defined for $\langle c|c+v=t \rangle$ and $\langle c|c+p=t' \rangle$. NO amplitudes in &cfp namelists are to be read in after this coupling.

The reverse case (KIND=12), where the projectile is composite rather than the target, is not yet implemented.

3.6.1 &CFP namelists

Spectroscopic amplitudes for the overlaps between partitions ICTO & ICFROM already defined by a &coupling namelist:

IN, IB, IA, KN, A

meaning that the overlap of the composite nucleus in excitation state IB with the core nucleus in excitation state IA is the bound-state form factor KN with amplitude A. The IN=1 specifies projectile overlap, and IN=2 target overlap.

If the form factor KN mixes different IA levels, then the spectroscopic amplitudes should rather be specified in an &overlap namelist.

N.B. The amplitudes A are signed, and are NOT the spectroscopic factors, but will typically be the square roots of these factors. For transfers out of or into closed shells of N antisymmetrised nucleons, the spectroscopic factors will usually contain factors of N , so the spectroscopic amplitudes needed by FRESCOX will typically need to already contain factors of $\sqrt{N}$.

The sign of A should be consistent with the spin coupling order used in the program, which is

$$(\ell, s)j, J_{core}; J_{com} \quad (12)$$

for binding a $\ell s j$ nucleon onto a core of spin J_{core} to form J_{com} .

If IN<0, use abs(IN) in this the last &cfp namelist

If IN=0, no more &cfp namelists to be read.

Glossary of Indexing Variables

IC, IC1, IC2 index to mass partitions

IA, IB index to projectile-target pairs of excited states within each partition

KP index to table of potentials, for scattering, binding, and folding

KN index to single-particle bound states

KN1,KN2 two-particle bound states are given for a range (RMIN to RNN) of internucleon separations, and are stored in the MSP table from KN1 to KN2 inclusive.

CP index to the coupling set up

4 Input CDCC format

&CDCC namelist

hcm, rmatch, rintp, hnl, rnl, centre, rsp, iter, pset, llmax, dry, rasym, accrcy, switch, ajswtch, sinjmax, cutl, cutr, cutc, absend, jtmin, jump, jbord, nnu, rmatr, nrbases, nrbmin, pralpha, pcon, meigs, listcc, smats, veff, chans, xstabl, thmin, thmax, thinc, smallchan, smallcoup, melfil, nosol, cdetr, numnode, treneg, nlpl, trans, pel, exl, cdccc, qscale, pade, kfus, elab, lab, lin, lex, hktarg, ncoul, reor, pauli, nk, q, ipc, iscgs, ipcgs, hat, remnant, postprior, quasi, sumform, qc, la, static, expand, maxcoup

are all the same as the **&FRESCOX** namelist, except for:

cdccc which is an alias for cdcc of &Fresco namelist,

q = ip1, multiple for the projectile single-particle couplings,

ncoul = ip2, selecting nuclear and/or coulomb,

reor = ip3, selecting diagonal and/or off-diagonal couplings,

qc = ip4, $Q_{\max}$ for the deformed core potential multipoles,

la = ip5, $\Lambda_{\max}$ for the new multipole orders with formfactor reduction,

hat (logical, default T) to use mean bin energies (otherwise midpoint),

quasi: if assigned, set all channel energies as if for this projectile energy (eg -be for adiabatic),

iscgs, ipcgs = isc, ipc (&overlap) for the projectile ground state wave function,

elab = ELAB (1), just the one projectile lab energy,

hktarg = target value of $h.K$, where $h=HCM$ and K is the elastic wave number. (default hktarg=0.2).

If HCM=0 in &Fresco namelist, then h is calculated from elab and hktarg.

sumform determines the bin formfactor reductions (default 2 if there are any kind=3 bin states, else 0):

= 0 : no formfactor reduction: $\langle KN|KQ\lambda|KNP \rangle$

= 1 : $KQ\lambda$ formfactors summed into new multipole Λ : $\langle KN|\Lambda|KNP \rangle$

= 2 : formfactors summed into composite projectile state: $\langle IB|\Lambda|IBP \rangle$. This is not allowed if have spin-orbit or transfer couplings, and sumform=1 will be set in these cases.

If have no cc bins, then K multipoles truncated using ip1=q.

If have cc bins and $Q_{\max}=qc=ip4$ and $\Lambda_{\max}=la=ip5$ are unset, then q sets the maximum multipole order of Λ , and all possible values of $KQ\lambda$ are used that couple to $\Lambda \leq q$

If qc and la are both set, then control each multipole individually:

q = $K_{\max}$, ip1 set in usual way

ip4=qc = $Q_{\max}$.

ip5=la = $\Lambda_{\max}$ maximum order for new multipole.

Note: λ will always run from $0 \rightarrow Q$.

trans determines the number of transfer partitions:

= 0 : no transfers: no E(jectile) or R(esidue)

< 0 : only R(esidue): the Ejectile and Core are identical

> 0 : both E(jectile) and R(esidue) independently of Core nucleus.

postprior = ip1 for finite-range transfer couplings,
 remnant = ip2 for finite-range transfer couplings,
 pauli = attempted Pauli blocking using non-orthogonality couplings.

&NUCLEUS namelist

part, name, mass, charge, spin, parity, be, n, l, j, ia, a, kind, lmax, nch, nce, ampl
 where this line is repeated for each part beginning P: projectile, C: core, V: valence, T: target, E: ejectile, R:
 residue. The number of nuclei (4, 5 or 6) depends on trans: see above.

name, mass, charge as name, mass, zp/t in a &partition namelist,

spin, parity as Jp/t, Bandp/t in a &states namelist.

be, n, l, j, ia, a, kind, lmax, nch prescribe the projectile P bound state, and with transfers also for
 residue R.

nce is the number of Core excited states.

If nce>0, then read that number of **&CORESTATES** namelists.

&CORESTATES namelist: spin, parity, ex

for spin, parity (± 1) and excitation energy of each Core state above the ground state.

&BIN namelist

spin, parity, step, start, end, energy, n, l, j, isc, ipc, kind, lmax, nch, ia, il, ampl

These are repeated until an empty **&BIN** namelist is encountered (step=0).

Each bin set has the same spin, parity, l, j, isc, ipc, kind, lmax, nch, ia, il, ampl,
 but a different energy. The energy range is divided into (end-start)/step bins. If energy then these are
 evenly spaced in energy, else they are evenly spaced in momentum $k \propto \sqrt{E}$. Changing or starting partial waves
 is equivalent to start=0.001.

The values of lmax, nch, ia, il are only needed coupled channels bins kind=3, with il being the
 channel number of the incoming partial wave, which, if il=0, is defined as the channel with quantum numbers
 l, j and ia. The array ampl gives overall multiplicative spectroscopic amplitudes to the bin. The values of
 isc, ipc, kind, lmax have the same meaning as in a &overlap namelist, with j an alias for jn.

If l, j, ia are not set then kind=3 coupled-channels bins are generated using lmax, for all incoming waves.
 Setting il will select an incoming channel.

&POTENTIAL namelist

part, a1, a2, rc, ac, v, vr0, a, w, wr0, aw, wd, wdr0, awd, vso, rso0, aso,
 shape, freal, fimag, vsot, rsot0, asot, l, parity, nosub, itt, beta2, beta3,
 ideo, beta2c, beta3c, beta2m, beta3m

where this namelist is repeated for each part beginning P: projectile-target optical potential, C: core-target
 optical potential, V: valence-target optical potential, T: projectile (C+V) ground state, B: projectile channels
not containing the ground state, T: transfer channel optical potential, E: ejectile bound state, R: residue bound
 state.

Different part=B(in) potentials may be defined depending on parity or partial wave l.

nosub means that the P(rojectile) optical potential is *added* to the CDCC couplings as a diagonal in all pro-
 jectile state channels.

For deformations, beta2m, beta3m are the nuclear *fractional* deformations, and beta2c, beta3c

are the Coulomb equivalents (both with default values `beta2`, `beta3` respectively).

The Coulomb and nuclear deformations may also be restricted by `idef`:

= 0 : Coulomb & nuclear (complex)

= 1 : nuclear (complex) only

= 2 : Coulomb only

Sample CDCC input file:

```
11Be+4He spdf; 1+5*10+2*5 chs 0-10 MeV, q=0-3 2200 MeV, 30/100 fm
CDCC
&CDCC
  hcm=0 rmatch=-30 absend=-50 rasym=100 accrcy=0.001
  elab=2200
  jbord= 0 60 200 2500
  jump = 4 5 20
  thmax=30 thinc=.05 smats=2 xstabl=1 cutr=-10 cutc=0
  nk=50 ncoul=0 reor=0 q=3
/
&NUCLEUS part='Proj' name='11Be' spin=0.5 parity=+1 be = 0.500 n=2 l=0 j=0.5 /
&NUCLEUS part='Core' name='10Be' charge=4 mass=10 /
&NUCLEUS part='Valence' name='neutron' charge=0 mass=1 spin=0.5/
&NUCLEUS part='Target' name='4He' charge=2 mass=4 /

&BIN spin=0.5 parity=+1 step=0.5 end=10. energy=F l=0 j=0.5/
&BIN spin=0.5 parity=-1 step=0.5 end=10. energy=F l=1 j=0.5/
&BIN spin=1.5 parity=-1 step=0.5 end=10. energy=F l=1 j=1.5/
&BIN spin=1.5 parity=+1 step=1.0 end=10. energy=F l=2 j=1.5/
&BIN spin=2.5 parity=+1 step=1.0 end=10. energy=F l=2 j=2.5/
&BIN spin=2.5 parity=-1 step=2.0 end=10. energy=F l=3 j=2.5/
&BIN spin=3.5 parity=-1 step=2.0 end=10. energy=F l=3 j=3.5/
&BIN /

&POTENTIAL part='Proj' a1=11 a2=4 rc=1.0 /
&POTENTIAL part='Core' a1=10 a2=4 rc=1.0
  V=46.92 vr0=1.204 a=0.53 W=23.46 wr0=1.328 aw=0.53 /
&POTENTIAL part='Valence' a1=4 rc=1.3
  V=37.14 vr0=1.17 a=0.75 W=8.12 wr0=1.26 aw=0.58 /
&POTENTIAL part='Gs' a1=10 v=51.51 vr0=1.39 a=.52 vso=0.38 rso0=1.39 aso=0.52/
&POTENTIAL part='Bi' a1=10 v=28.38 vr0=1.39 a=.52 vso=0.38 rso0=1.39 aso=0.52/
```

5 Input SFRESCOX format

This is an program which search for a χ^2 minimum when comparing the outputs of FRESCOX with sets of data, using the MINUIT search routines.

The inputs for SFRESCOX specify the FRESCOX input and output files, the number and types of search variables, and the experimental data sets.

5.1 Search file

A `search_file` format is defined as follows:

```
(2*string, 2*integer: free format)
input_file, output_file, nvariables, ndatasets
```

`input_file`: name of file (in single quotes) with regular FRESCOX input
`output_file`: name of file for the final FRESCOX output
`nvariables`: number of search variables
`ndatasets`: number of experimental data sets.

Repeated `nvariables` times, a namelist with some of the following variables:

```
namelist &variable:
name, kind, step, valmin, valmax, kp, pline, col, potential, nopot,
dataset, datanorm, dataEshift, reffile, nafrac, afrac,
energy, damp, jtot, par, channel, width, rwa, B, leff,
pline2, col2, ratio2
```

`name` is name of variable (up to 10 characters),
`kind`: 0=ignore, 1=potential, 2=afrac, 3=R-matrix energy,
4=R-mat partial width, 5=dataset normalisation
`step`: step size for finding derivatives etc, zero for fixed. (default 0.01)
`valmin`: strict lower bound (if non-zero),
`valmax`: strict upper bound (if non-zero),
`null`: (default -124578) 'undefined' value,

`kind=1`: **Variable potential parameter:**
`kp`: number of potential as in `&pot` namelist,
`pline`: order of which `&pot` namelist within potential definition,
`col`: number within namelist: selecting P0–P7 ,
`potential`: value of this potential parameter (default `null`: use `input_file`)
`pline2, col2`: if both assigned, then potential parameter at line `pline2` and index `col2` is automatically set to a value of `ratio2` times this parameter's value.

kind=2: Variable spectroscopic amplitude:

nafrac: order number of &cfp namelist in input_file
afrac: spectroscopic amplitude A (default null: use input_file)

kind=3: Variable energy of R-matrix term:

term: term number (default 1)
jtot: J value for additional R-matrix term
par: parity (+1 or -1)
energy: energy (cm MeV in entrance channel) of additional R-matrix term (default 0)
damp: energy (cm MeV in entrance channel) of imaginary width of this R-matrix term (default 0). The actual imaginary part is damp/2
nopot: Disable potential & Buttle correction for this J/pi set (logical T or F, default F)

kind=4: Variable partial width of R-matrix term:

channel: channel number (in order generated by FRESCOX)
term: term number (default 1)
rwa: the width is reduced width amplitude γ , else $\Gamma = 2\gamma^2 P$. (default T)
B: the boundary condition B. (Default: from BNDX).
width: width for this channel (MeV^{1/2} if rwa or energy subthreshold, else MeV(cm)) (default 0)

kind=5: Variable dataset normalisation:

dataset (1:2): index (or, first and last, inclusive) of which dataset (1 to ndatasets) (default 1), or
reffile: filename of dataset. If has a '*', renorm all sets with names matching up to that character,
datanorm: search on absolute normalisation factor of this dataset and its error (default: 1.0)

kind=6: Variable dataset energy shift:

dataset (1:2): index (or, first and last, inclusive) of which dataset (1 to ndatasets) (default 1), or
reffile: filename of dataset. If has a '*', shift all sets with names matching up to that character,
dataEshift: search on absolute energy incremental shift of this dataset (default: 0.0)

kind=7: Variable damping width of R-matrix term:

term: term (default 1). For the real energy there should be a kind=3 variable with the same term.
damp: Imaginary width (cm MeV in entrance channel) of this R-matrix term (default 0).
When energy is set, use the (real) l_{eff} to rescale the damping width according to energy above threshold energy by the penetrability ratio $P_{\text{leff}}(E - \text{energy})/P_{\text{leff}}(e_{\text{pole}} - \text{energy})$, with elastic Coulomb. If not set, use constant damping widths as 2*imaginary parts of the pole energies.
The damp variable in kind=3 is fixed, but those in kind=7 can be varied in searches.

Repeated ndatasets times, namelist with some of the following variables, then the corresponding data:

```
namelist \&data:
type,data_file,points,delta,xmin,lab,energy,angle,
idir,yscale,abserr,ic,ia,k,q,jtot,par,channel,value,error,leg,ib,
pel,exl,labe,lin,lex,term
```


`type` (default 0)
 = -3 Legendre coefficient for input energy and fixed order `leg`
 = -2 Legendre coefficient (integer order `leg`) for input energy: double distribution
 = -1 Legendre coefficient for varying integer order `leg`, for fixed energy
 = 0 angular distribution for fixed energy
 = 1 excitation and angular cross section double distributions
 = 2 excitation cross section for fixed angle
 = 3 excitation total cross section. Also: `ic=0`: `ia=0` is total reaction cross section; `ia=1` is total fusion cross section; $1 < ia < NFUS+1$ are fusion from potential KFUS (in &Fresco namelist) for inelastic state `ia-1`.
 = 4 excitation phase shift for fixed partial wave
 = 5 desired factor for bound state search (binding energy or potential scale according to ISC).
 = 6 specific experimental constraint on a search parameter `par`, to be value with error `error` (`abserr=T` or `F`).
 = 7 specific constraint on energy of R-matrix pole term in the Brune basis to be value with error `error` (`abserr=T` or `F`).
 = 8 specific constraint on total formal width of R-matrix pole term in the Brune basis to be value with error `error` (`abserr=T` or `F`).

`data_file`: name of data file with data, '=' for `search_file`, '<' for stdin (default '=')
`points`: number of data points (default: keep reading as many as possible),
`delta`: if non-zero, construct linear x-scale from `xmin` in steps of `delta`, (default 0)
`lab`: T for lab angles and cross sections (default F for cm frame).
`Aflip`: T to change cm calculated angles to $180 - \theta$ (default F).
`energy`: lab energy for this `type=0` dataset (default: use `ELAB(1)` from &Fresco namelist)
`pel, exl, labe, lin, lex`: specify input channels for this data set.

`idir` (default 0)
 = -1 cross-section data are given as astrophysical S-factors, but will be converted to absolute,
 = 0 cross-section data are given in absolute units,
 = 1 cross-section data are ratio to Rutherford
 = 2 cross sections are given in absolute units but will be converted to ratio to rutherford.

`iscale` (default 2)
 = -1 dimensionless (eg ratio to rutherford if `idir=1`),
 = 0 absolute cross-section units are fermi-squared/sr.
 = 1 absolute scale is barn/sr
 = 2 absolute scale is mb/sr
 = 3 absolute scale is micro-b/sr

`abserr`: T or F for absolute error (default F)
`ic`: partition number of cross section channel (index of Cards 6) (default 1)
`ia`: excitation pair number of cross section (index of Cards 7) (default 1)
`k`: tensor rank of cross section (0, 1, 2,...) (default 0).
`q`: index for tensor T_{kq} of cross section (0, 1, 2,...) (default 0). $T_{00} = \text{angular distribution } \sigma(\theta)$.
`jtot`: The J value (as on Card 2) for phase shift
`par`: The parity (-1 or +1) for phase shift, for `type=4`
`par` or number of search parameter, for `type=6`.
`channel`: Channel number of entrance channel in coupled channels set (default 1), for phase shift

value: Value of search parameter, for type=6
error: Error in value of search parameter, for type=6

ib: if $0 < ib < ia$, then ib is final state of gamma decay. (default 0)

If this is given for data type=0,1,2, then the ia is the initial state, and all angles are the *gamma* detection angles, and cross sections are integrated over all ejectile particle angles. Need to set dgam=1, pp=3, kqmax $\geq 2 \times$ spin of state ia, thmin=0 and thmax=180 degrees.

Data input (free format):

if type=1 or -2, read: energy, angle, value, error
if type=5, read: kn, target, error
otherwise:
if delta $\neq 0$, read: value, error
if delta=0, read: x, value, error

where x is angle for type=0, is leg for type=-1 or -2, and lab energy for type=-3, 2, 3 and 4.

5.2 Sfrescox commands

The inputs to SFresco are

Card 1: name of search_file

Lines 2: repeatedly one of (in all upper or all lower case):

EX: exit

Q: query status of search variables

V: query status of unfixed search variables

CHA: query status of changed search variables

CHI: list χ^2 status by dataset

SET var val: set variable number var to value val.

FIX var: fix variable number var (set step=0).

FXWID : fix all R-matrix widths variables.

FXRPE : fix all R-matrix pole energy variables.

FXPBG : fix all pole energies variables with 'BG' in their name.

FXBG : fix all variables with 'BG' in their name.

FXNOR : fix all dataset norm variables (kind=5)

FXSHF : fix all dataset energy shift variables (kind=6)

FREE : free all R-matrix energies and widths variables.

STEP var step: unfix variable number var with step step.

SCAN var val1 val2 step: scan variable number var from value val1 to value val2 in steps of step.

SHOW: list all datasets with current predictions and χ values.

PLOT plotfile: write file (default: search.plot) for reading by xmgr, xvgr, with data and theoretical curves.

DONE search_file: write single new search file (default: done.fresco) for rereading by SFRESKO as a combination of FRESKO input_file and current parameter values. Data specifications are however not included.

LINE plotfile: write file (default: search.plot) for reading by xmgr, xvgr, with theoretical curves only.

READ file: read plot output file for further searches, if not:

READ snapfile: if the input string file contains the string 'snap', read to the end of previous snap output snapfile, for further searches.

ELIM emin emax: restrict data scans to this energy range for plots and χ^2 sums

POLES emin emax: vary only R-matrix poles in this energy range (others are fixed)

ESCAN emin emax estep: scan lab. energy in incident channel, to give phase shifts in fort.71, fusion & reaction cross sections in fort.40, and S-factors in fort.35 and 75. This is to look for/at resonances, or at behaviour out of data's energy range, not for χ^2 fitting.

MIN: call minuit interactively, to read from stdin. Return to SFresco by minuit command end.

A example of this search_file input is:

```
'6hep.in' '6hep.out' 7 1
&variable kind=1 name='V' kp=1 pline=2 col=1 /
&variable kind=1 name='r0' kp=1 pline=2 col=2 valmin=.7 valmax=2.0/
&variable kind=1 name='a' kp=1 pline=2 col=3 /
&variable kind=1 name='W' kp=1 pline=2 col=4 /
&variable kind=1 name='rw' kp=1 pline=2 col=5 valmin=.7 valmax=2.0/
&variable kind=1 name='aw' kp=1 pline=2 col=6 /
&variable kind=5 name='exptnorm' /
&data type=0 iscale=2 idir=0 lab=F abserr=T/
13.99      565.7      28.3
17.59      478.19     23.91
21.22      425.41     21.28
24.92      338.44     16.93
28.68      284.71     14.24
36.48      168.59      8.43
40.56      137.9       6.86
44.8       104.75      5.24
53.        30.0        9.14
53.94      38.45       2.71
59.        25.62       2.55
&
```

A example of the full SFresco input, suitable for batch running, is:

```
search4.in
min
fix 4
migrad
```

```
end  
q  
show  
plot
```

where `search4.in` is the above `search_file` input file name.

6 Test run: a CCBA calculation

The following test case calculates the inelastic and transfer cross sections for the reactions induced by ^{19}F on ^{28}Si at 60.0 MeV. A Triton-Oxygen cluster structure is assumed for ^{19}F , with the potentials for the triton and the oxygen core being folded together to give a diagonal Watanabe potential for elastic ^{19}F scattering as well as the couplings to the inelastic $1/2+$ state. These elastic and inelastic channels are coupled together to all orders, then a finite-range transfer step is calculated for the $^{16}\text{O} + ^{31}\text{P}$ channel, thus performing a CCBA calculation.

Test run: CCBA calculation 28Si(19F,16O) 31P, cluster form factors.

```

NAMELIST
&FRESKO
hcm=.10 rmatch=25 rintp=0.50 hnl=0.100 rnl=3.
jtmin=0. jtmax=80 absend=.01 kqmax=1 thmin=0. thmax=60.0 thinc=2.5
iter=1 iblock=2 nnu=30 chans=1 listcc=2 smats=1 elab=60.0
/
&PARTITION
  namep='19-F' massp=19. zp=9 nex=2 namet='28-SI' masst=28 zt=14 /
&states Jp=0.5 Bandp=+1 Ep=0.0 cpot=1 Jt=0.0 bandt=+1 Et=0.0 /
&states Jp=2.5 Bandp=+1 Ep=0.2 /

&PARTITION
  namep='16-O' massp=16. zp=8 nex=1 namet='31-P' masst=31 zt=15 qval=6.199/
&states Jp=0.0 Bandp=+1 Ep=0.0 cpot=2 Jt=0.5 bandt=+1 Et=0.0 /
&partition /

&pot kp=1 At=28.0 Ap=19.0 Rc=1.2 /
&pot kp=2 At=31.0 Ap=16.0 Rc=1.35
  V=31.20 rv=1.45 av=0.47 W=15.10 rw=1.270 aw=0.310
  Vso=0.75 rso=1.24 aso=0.37 /
&pot kp=3 At=19.0 Rc=1.25 ac=0.65
  V=115.00 rv=1.250 av=0.650 Vso=6.30 rso=1.25 aso=0.650 /
&pot kp=4 At=31.0 Rc=1.25 ac=0.65 V=99.000 rv=1.250 av=0.650
  Vso=6.30 rso=1.25 aso=0.650 /
&pot kp=5 At=28.0 Rc=1.2
  V=27.00 rv=2.42452 av=0.460 W=11.66 rw=2.14456 aw=0.238 /
&pot kp=6 At=28.0 Rc=1.25 ac=0.65
  V=160.00 rv=1.07 av=0.72 W=37.0 rw=1.35 aw=0.88
  Vso=10.0 rso=0.85 aso=0.850 /

&pot /
&OVERLAP kn1=1 ic1=1 ic2=2 in=1 kind=0 nn=4 l=0 sn=.5 j=0.5 kbpot=3 be=11.73 isc=1 /
&OVERLAP kn1=2 ic1=1 ic2=2 in=2 kind=0 nn=4 l=0 sn=.5 j=0.5 kbpot=4 be=17.899 isc=1 /
&OVERLAP kn1=3 ic1=1 ic2=2 in=1 kind=0 nn=3 l=2 sn=.5 j=2.5 kbpot=3 be=11.53 isc=1 /
&overlap /
&COUPLING icto=2 icfrom=1 kind=7 ip1=0 ip2=0 ip3=0/
&cfp in=1 ib=1 ia=1 kn=1 a=1.00 /
&cfp in=1 ib=2 ia=1 kn=3 a=1.00 /
&cfp in=2 ib=1 ia=1 kn=2 a=1.00 /
&cfp /
&COUPLING icto=1 icfrom=2 kind=3 ip1=4 ip2=1 ip3=0 kfrag=6 kcore=5/

```

7 File allocations

File no.	Format (Fix/Var)	LRECL (bytes)	Common suffix	Routines	Use
1	V			SFRESCOX	FRESCOX input when searching
2	V			SFRESCO	Search specification file
3	V			FREADF, FR	Temporary namelist file
4	F	80		INTER POTENT	default for external KIND=1,9 couplings external potentials
5	F	80	in		standard input
6	V	133	out		standard output
7	F	80	smats	DISPX	elastic S-matrix elements
8 #	F	sector	(tmp)	FR,INTER	s/p wfs, channel wfs
9 #	F	NLL*NLO*16	(tmp)	Q/KERNEL	complex transfer multipoles
10 @	V		(tmp)	FR,CRISS	S-matrix elements (cfs)
11 #	F	NLL*NLO*8	(tmp)	Q/KERNEL	real transfer multipole
12 #	V	NLL*NLO*16	(tmp)	KERNEL/SOURCE	transfer kernels
13	F	80	xst	FR	total cross sections for each Elab
14 #	V			INTER/CPAIR	interaction potentials
15	V		(tmp)	FR	local equivalent potent
16	F	80	xsec	CRISS	tables of cross sections
17	F	80	swf	FR	output scattering waves
18 #	V		(tmp)	FR	wfns of 'best' iterate
19	F	N		FR	Local couplings
20-33				Available for users	(eg bound states, amplitudes, potentials)
34	F	80		POTENT	output potentials
35	F	80	sf	FR	Astrophysical S-factors for Ecm
36				CRISS	output scattering AMPL amplitudes
37				CRISS	output scattering FAM amplitudes
38	F	80		DISPX	cross sections for each J/pi
39	F	80	efus	FR	2 cross sections for each Elab
40	F	80	fus	FR	all cross sections for each Elab
41	F	80		SOURCE	source terms at each iteration
42	F	80		SOURCE	bin wavefunctions for each E
43	F	80		INFORM	bin phase shifts as k functions
44	F	80	phe	INFORM	bin phase shifts as E functions
45	F	80		ERWIN	scat phase shift as E functions
46	F	80		INFORM	bs wave functions & Whit ratios
47	F	80	mek		reduced matrix elements
48 #	V	133		FR	concurrency log file
49 #	V			FR,CRISS	S-matrix elements concurrent→10

File no.	Format (Fix/Var)	LRECL (bytes)	Access (Seq/Ran)	Routines	Use
50 #	V	133			tmp standard output concurrent→6
51,52 #	V	133			unwanted standard output for node N>0
53	V		mel	WRITEMEL	mel file for sturmx(x)
54	V		spec	WRITESPEC	spec file for sturmx(x)
55	F			INFORM	Single-particle wave functions
56	F		fusl	FR	Jtotal fusion & reaction cross
57	F		cdcc	FR	Output of CDCC amplitudes
58	F		wfs	INFORM	Bound state wave functions
59	F		vrt	INFORM	Bound state vertex functions
60-62	F		imp	RMATRIX	Trace of R-matrix calculations
63	F			WRITEMEL	Asymptotic coupling coefficients
64	F			RMATRIX	Coupled eigenstates
65 #	F	NLO*2*8	(tmp)	QNN	two-particle multipoles
66	V	NLL*NLO*16		INTER	KIND=1 nonlocal formfactor
70	F			RMATRIX	Basis state wfs
71	F	132	phs	FR	20 pw phase shifts as Elab functions
79+CP	V	{N,NLN*NLO}*16	(tmp)	INTER/KERNEL	(non)local partial-wave couplings
87	V		pnl	KERNEL	Non-local couplings
88	F			RMATRIX	Bound state wfs
89	F			MULTIP	Folded coupling potentials
90	V		veff	FR	average local equivalent potential
91	V		vpol	FR	average LEP for rereadingl
92	F			FR	Partial fusion cross sections
99	F	80		CRISS	input J/L rescaling factors
105	V			FCN	χ^2 progress during fitting
200	F	80	xsel	CRISS	Elastic cross section if not fort.201
201-210	F	80	xs[1-9]	CRISS	Cross sections (cf 14) for states 1-10
301	V			CDCIN	New Fresco input
302	V			CDCIN	New Fresco input (temp)
303	V			SFRESKO	Input search file
304	V			SFRESKO	Output plot file
305	V			CDCIN	New Fresco input from cdcin, col format
306	V			SFRESKO/FRXX0	Input Fresco file
307	V			SFRESKO/FRXX0	Initial Output Fresco file
308	V			SFRESKO/FRXX0	Main Output Fresco file
470+i	V			erwinmpi	Log files from MPI process <i>i</i>

where for MACH=8 we have

= distributed file: distinct file name on each node

@ = shared concurrent file: single file, with writes from each node

The use of files 19 and 35 have recently changed.

8 KIND=1 Spin Transfer Couplings

The definition of the KIND=1 spin transfer couplings is not given in the Computer Physics Reports article, so in versions after March 1998 these are redefined for IP3=0 or 1, and new Racah algebra factors included.

We want to calculate the coupling interactions of the monopole operator $\mathbf{S}([\ell, s_p]s_t, s_t)$, where s_p is the spin transfer of the projectile I_p , s_t is the spin transfer of the target I_t , and ℓ is the orbital angular momentum transfer. These coupled operators are defined following in Bohr & Mottelson, Vol. 1, section 1A-5c, as

$$(F_{\lambda_1} G_{\lambda_2})_{\lambda\mu} = \sum_{\mu_1 \mu_2} \langle \lambda_1 \mu_1, \lambda_2 \mu_2 | \lambda \mu \rangle F_{\lambda_1 \mu_1} G_{\lambda_2 \mu_2}, \quad (13)$$

applied for the case of $F_{s_t} = [\ell, s_p]s_t$ and $G_{s_t} = s_t$. The overall $\mathbf{S}$ operator is a monopole (scalar), for which the tensor product is thus of the kind

$$(F_{\lambda} G_{\lambda})_{00} = (2\lambda + 1)^{-1/2} \sum_{\mu} (-1)^{\lambda-\mu} F_{\lambda\mu} G_{\lambda-\mu}. \quad (14)$$

This differs from common definitions (eg of Satchler) by a factor of $(-1)^{\lambda}(2\lambda + 1)^{-1/2}$.

Reduced matrix elements are defined everywhere in FRESKO by:

$$\langle j_f m_f | \hat{O}_{\lambda\mu} | j_i m_i \rangle = (2j_f + 1)^{-1/2} \langle j_i m_i, \lambda \mu | j_f m_f \rangle \langle j_f || \hat{O}_{\lambda} || j_i \rangle. \quad (15)$$

The reduced matrix of a (LOP=0) spherical harmonic $\mathbf{S}(\ell) = Y_{\ell}(\hat{\mathbf{R}})$ is

$$\langle L' || Y_{\ell} || L \rangle = \frac{\hat{\ell} \hat{L}}{\sqrt{4\pi}} \langle L0, \ell 0 | L'0 \rangle \quad (16)$$

and the reduced matrix of a (LOP=1) vector angular momentum operator $\mathbf{S}(\ell) = \ell$ itself is, for $\lambda = 1$ vectors,

$$\langle L' || \ell_{\lambda} || L \rangle = \delta_{LL'} (-1)^{\lambda} \sqrt{2\lambda + 1} \sqrt{L(L+1)(2L+1)}. \quad (17)$$

A spin-orbit force, for example, must use LOP=1 with $\lambda = 1$, since eq(16) gives zero in this case for diagonal interactions. This will use $\ell \equiv \text{LTR} = 1$ and $s_p \equiv \text{PTR} = 1$ in the data file for the KIND=1 input. There is not yet the coding for off-diagonal spin-orbit forces from e.g. deformations.

The matrix elements of the full coupling operator are

$$\begin{aligned} & \langle (L I_p) J, I_t; J_T M_T | \mathbf{S}([\ell, s_p]s_t, s_t) | (L' I'_p) J', I'_t; J_T M_T \rangle. \\ &= (-1)^{s_t + J_T + J' + I_t} \left\{ \begin{matrix} J' & I'_t & J_T \\ I_t & J & s_t \end{matrix} \right\} \hat{J} \hat{J}' \left\{ \begin{matrix} L' & I'_p & J' \\ \ell & s_p & s_t \\ L & I_p & J \end{matrix} \right\} \langle L' || \mathbf{S}(\ell) || L \rangle \\ & \quad \times \langle I_p || s_p || I'_p \rangle \langle I_t || s_t || I'_t \rangle \end{aligned} \quad (18)$$

In using KIND=1 couplings :

With IP3=0, the first line of these factors is generated automatically. The product of the reduced matrix elements for the intrinsic nuclear states, $\langle I_p || s_p || I'_p \rangle \langle I_t || s_t || I'_t \rangle$, has to be included explicitly in the factor FSCALE, or in the radial form factors. These reduced matrix elements will be from eq(16) or eq(17). The radial shapes are to be read in from data files.

With IP3=2, implying jlmP input for target-only couplings,

- the projectile diagonal reduced matrix element $\langle I_p || s_p = 0 || I_p \rangle = \hat{I}_p$ is supplied by Fresco,
- the factor $1/\sqrt{4\pi}$ is omitted above,
- and an additional symmetric factor of $((2I'_t + 1)(2I_t + 1))^{1/4}$ is supplied to allow the monopole radial form factors to have their physical values for any target spin.
- The forward and reverse form factors should be identical for Hermitian couplings.

With IP3=3, implying a similar input for projectile-only couplings,

- the target diagonal reduced matrix element $\langle I_t || s_t = 0 || I_t \rangle = \hat{I}_t$ is supplied by Fresco,
- the factor $1/\sqrt{4\pi}$ is omitted above,
- and an additional symmetric factor of $((2I'_t + 1)(2I_t + 1))^{1/4}$ is supplied to allow the monopole radial form factors to have their physical values for any target spin. The forward and reverse form factors should be identical for Hermitian couplings.